



Mendel University in Brno

Faculty of Forestry and Wood Technology

Department of Furniture, Furniture Design and Habitation

Sustainability in Furniture design

DISSERTATION THESIS

with appendices

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two hundred testing samples were created, and their physical and mechanical properties were tested. Also, its influence on the environment with the VOCs were tested. The research started during the internship at Louisiana State University (LSU), continued at Salzburg University of Applied Sciences (SUAS) and continued at Mendel University in Brno (MENDELU). During the work on the thesis the author also cooperated with a company Plastia, based in Nové Veselí, Czech Republic. The company provided material for further testing. The boards with the best results were used for furniture design of coffee tables and info panel, which was exhibited during the biggest Design event in the Czech Republic, Designblok 2020.

Key words

paper, pulp, board, composite, post-consumer, post-factory, furniture, design, sustainability

jsou desky, které byly použity na výrobu odkládacích stolků a dále pak info panelu pro výstavní účely. Výsledek práce byl prezentován v rámci výstavy Designblok 2020 v Praze.

Klíčová slova

papír, papírovina, desky, kompozit, post-consumer, post-factory, nábytek, design, udržitelnost

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1. Introduction

Demands on the ecology and sustainability has increased in the last years. We live in a time of countless options of different products from several different materials. According to Ashby et al. (2007), there is over 160 000 materials over the whole world. Packaging materials such as polyethylene bags, polystyrene lunch boxes and others are constantly contributing to higher the pollution of the Earth. Companies are producing a lot of products and trying to earn as much money as they can. Many times, there are born products, which has only aesthetical purpose and they are losing its lifetime during a few years. Americans are the 5 % of the world's population but they are using 25 % of the world's energy. Their production belongs to one of the biggest in the world. In ideal world the products would be manufactured with long lifetime which lowers the environmental impact (Holman, 2015). Consumption of materials in the USA is exceeding 10 tonnes per person every year. The resources of materials on our Earth are not infinite but mankind did not realize that until some years ago (Ashby, 2009).

Even in the primitive ages, human race was smart enough to create primitive furniture in their caves, which they used for their living. Over time, human have realized that they can use wood as easy to work with material for furniture. The great manufacture has begun and increased over years. Other materials were developed as well, which overtime has led to a big ecological problem with non-disputable impact to the Earth (Uysal, 2014).

The goal of this dissertation thesis is to manufacture paper-based boards and use them in furniture design while reusing wastepaper and upcycle it into another functional product. The research has started with the use of recycled paper pulp such as deinked pulp (DIP) and old corrugated cardboard (OCC) with starch (S) and urea formaldehyde (UF) glue. These paperboards were manufactured at different temperatures to find out the best option. The mechanical and physical properties of the manufactured boards were tested.

The results were used to evaluate the best options and were also used to find another option for binding the paper-based boards.

Based on the previous research of the possibilities which were either different types of glue or even plastic materials, other trials were done with bioplastic polylactic acid (PLA). There was the idea of using plastic granulate combined which was used in a similar way like in the case of industrially produced wood plastic composite (WPC) boards. Further investigation brought an idea of using PLA granulate to get eco-friendly composite boards for the use in furniture design. PLA is a biodegradable material, although it needs to be placed under special composting conditions to decompose (Haider, et. al, 2018). In further development of this research, the used paper was recycled cardboard (CB) and used office paper (OP), where the milling was done by a high energy milling machine (HEM). This material was then pressed, and the paper-PLA boards were manufactured.

Fortunately the new trend is to know the origin of a product and used materials itself and the people are more willing to help the Earth with ecological lifestyle (Jonášová, 2021). In case of furniture, its lifetime is about 5 to 10 years and usually it is thrown away after the end of its lifetime. Even though wood, for example, is natural and ecological materials, it would take about 13 years to biodegrade in the landfill (Uysal, 2014).

Important terms in the production are eco-design and sustainability. What do these words mean? Eco-design means thinking about the product in its whole function, aesthetics, and ecology. A designer should be able to design not just the form of the product but also change the manufacturing process and its sustainability. Sustainable products can include furniture from post-consumers and post-factory materials also the upcycling of already used products or products made of biodegradable materials (Barbero, et al., 2012). The environmental impact of products is influenced during the design stage already. That is why furniture and product designers should think about the life cycle of their products while creating the new product (Thackara, 2007).

Available technologies also help the reduction of used materials thanks to modern technologies.

Many researchers are looking for a way of creating new composite materials which are manufactured from ecological and inexpensive materials. One of the ways is to reuse bio-based materials and waste materials. Bio-based materials are being used and tested as substitute for other materials and they are being manufactured and used in furniture design applications. For example, raw materials, such as banana stems, rice straw, bamboo or coconut husks are available materials around the world, and they can be used in furniture design. Their physical and mechanical properties are similar to wood, and they are suitable for wood-based panels (Surip et al., 2012). In Egypt, there is generated about 30 million tons of agricultural waste every year. This material can also be reused in a sustainable way for production of paperboard material (Elbasiouny, et al., 2019). Recycling agricultural waste can also help to reduce the amount of emitted carbon dioxide, as these materials are utilized rather than burned. Additionally, bio based medium-density fiberboards (MDF) and particleboards made from straw and bagasse tend to have better physical and mechanical properties, such as internal bond (IB), modulus of rupture (MOR) and modulus of elasticity (MOE), swelling in thickness and water absorption (WA), and screw withdrawal resistance (SWR), compared to traditional wood-based boards. They are also non-harmful to the health of the users because formaldehyde-based resins are not used in their production (Shehata, 2016).

Also Czech companies are thinking about the problem with waste materials and sustainability. Many of them have started to use waste material in their products too. One of them is successful Czech company "mmcité", which is using materials called Resysta. Resysta is an ecological material consisting of rice skins, natural oil, and stone salt. Composite materials are providing designers their quality and long-life span. In comparison with traditional materials, composites are developed with higher strength, lower weight, and/or for less maintenance. Recycling and reducing the waste can

contribute to better wellbeing of the planet – lower the greenhouse gasses (Chung et al., 2019).

Also, the used technologies are important in the sustainable process. Modern technologies such as CNC milling, laser cutting and 3d printing, can be considered sustainable in some ways. For example, 3d printing, which is not only regular Fused Deposition Modeling (FDM) but also other 3d printing technologies, which can create end-use products. Computer modelled designs can be printed or manufactured without scrap materials in comparison with the traditional way of manufacturing products where there can be a lot of waste produced. It is done by using local supplies and there is no need for mass production. It is possible to 3d print metals, sand, plastic, food and even life cells. In the design process it can help with creating prototypes which are easy to re-adjust, but in several areas such as automotive or aerospace design 3d printing of end-use parts is commonly used. This technology was also very important for the whole research process as a starting point to find sustainable way of designing furniture, because it started with the idea of 3d printing paper powder (Rael, et al., 2018).

Paper is an attractive material thanks to its life-cycle analyses: it is a fibrous material that is renewable, manufactured on a large scale, it is also light and recyclable. The recycling of paper contributes to the wellbeing of the environment and helps to conserve natural resources and it can be used for creating new products (Madyira, et al., 2017). There is a big amount of paper's consumption and the lack of recycling is an environmental problem as well. Even though paper can be recycled, its fibres are lose their strength each time. Also there is a problem with the technological process. Paper has to be bleached, the ink must be gotten rid of and other additives which have a negative impact on the environment as well (Dukarska, et al., 2018).

Although, waste paper is a suitable material that can be used as an additional material instead of wood to produce fiberboard panels in a similar way as particleboards or MDF. The author focused the most on this

problematic during the research. Paper and paperboard material is very often used as a packaging material for folding boxes but also plates or other kitchen appliances. For example, paper with inferior quality can still be reused for egg trays and are molded over a form (Groche, et al., 2016). Paper is a cellulose based material, which consists of wood pulp fibers, which are formed over a mesh-like screen. There the pulp fibers are dispersed in water at a solid consistency. The structure of paper is usually bonded by adhesives such as starch and the properties of a sheet of paper can be modified by using different composition in order to change the porosity or other structural properties (Xia, 2002). Even though paper is mostly sorted out and recycled, the amount of it is so big that it is not possible for all of it to be used (Dukarska, et al., 2018).

The need of environmentally friendly products should go hand in hand together with circular economy principles. The main idea of the circular economy is to give other functions to already used products or to use them as a material for new products with higher qualities. This process is called upcycling. Thanks to this process, people should be using waste material, which is a great challenge for every designer. Many companies are already working with the idea of the circular economy in furniture design. Several companies are either using waste material for creating their furniture or giving some product new function (Jonášová, 2021). The incorporation of recycled and waste materials into the manufacturing process can improve the environmental impact of the products.

The main goal of the research was to focus on reusing salvaged materials but also upcycling old products or materials. All of these goals were tested or tried during this research. It started with the cooperation of different companies, who provided salvaged materials or products for the International Design Workshop which the author helped to organize. Then she also started with her own project of manufacturing and testing board materials, which could be used in furniture design. She started with the reusing of waste paper material for the production of furniture while researching the options of technology that could be used for the manufacturing of furniture. Many options

were considered. It started with the idea of 3d printing paper and finally it ended with manufacturing the boards in a similar way as industrially manufactured wood-based products. Finally, the paper-based boards were manufactured and tested. The paper-based boards which were created during the research could theoretically be broken down into smaller pieces and upcycled again as another material for pressing the new boards, although it was not tested yet. This process is similar to the manufacturing process of particleboards, which are also being recycled.

The goal of this thesis was not to develop a substitutional material for wood-based products and go into depth with chemistry, but explore other options for a designer while designing new products in a sustainable way. She focused on reusing salvaged paper while using it in furniture design for a specific purpose described in this work.

2. Goal

The goal of this work is to describe the process of designing new products while taking to mind sustainability and its environmental impact on the environment. The aim is also to increase the awareness of the sustainability to furniture designers while designing new products and bringing some more suitable solutions for their products and designs while creating them. It all starts with research on this topic, finding sustainable materials and products which are developed with the idea of sustainability and the circular economy.

The main objectives of the thesis are:

- Map the current situation in furniture design in terms of sustainability.
- Increase the awareness of the sustainability in furniture design.
- Show how to reuse old products and give them another value.
- Demonstrate how to reuse salvaged materials for new products in cooperation with a company.
- Determine the strength of new composite materials made from salvaged materials.
- Describe the process of designing new products in a sustainable way.
- Present the results at a conference and exhibition, then process the feedback.

Another goal of this dissertation thesis is to investigate the possibility of reusing post-consumer wastepaper and cardboard and develop paper-based boards, which can be used in a similar way as other industrially manufactured wood-based products such as MDF or particleboards. The goal was to manufacture the boards by using a hydraulic pressing machine and test its physical and mechanical properties which was done according to the most

important properties for such wood-based products used in furniture industry.

The whole research process was combined with several experiences from abroad, which had a great impact on the whole thesis. Also, the upcoming demands on sustainability in all areas of human life brought the author to this topic.

The researcher has one-year teaching experience at Louisiana State University (LSU) in Baton Rouge, Louisiana, USA. The author of this thesis was teaching 3d modeling in 3ds Max and co-teaching Interior Component Design, where the main topic was nomadic lifestyle furniture. Part of the work was to organize international workshop which was held at Mendel University in Brno at the Castle Křtiny. American students worked together with Czech students of design on the chosen topic. The outcome of the three-day workshop were different design ideas on the topic which were also manufactured by students themselves at LSU after their arrival. The researcher organized an exhibition at the company Monochrome in Baton Rouge, Louisiana, where the students work was exhibited at the end of semester.

During that time, the author started to use 3d printing - Fused Deposition Modeling (FDM) and Stereolithography (SLA) 3d printers for printing work of students. While working there, the author researched about different materials other than plastic filaments for 3d printing such as saw dust, salt, coffee grounds, ceramics, concrete but also rubber and paper. From that there came an idea about 3d printing with paper although this technology is quite expensive, and it would be very difficult to continue with the development and research with provided technologies in the Czech Republic. The process of work continued with the idea of using paper for further development, only by using different technology – pressing board material in a similar way of manufacturing MDF boards. During that time, the author started to experiment with paper pulp together with a researcher Leah Hamel for functional objects.

First trials with the paper started while collecting cardboard or paper cartons trays used for fruit in grocery stores. The paper was cut into smaller

pieces, blended, and mixed with water. This is how the recycled paper pulp is made. Other materials can be added into the mixture, such as different types of fabric or other plant fibers can be cooked and used for making paper pulp. Handmade paper was made, and small mold casts were produced (Figure 1).

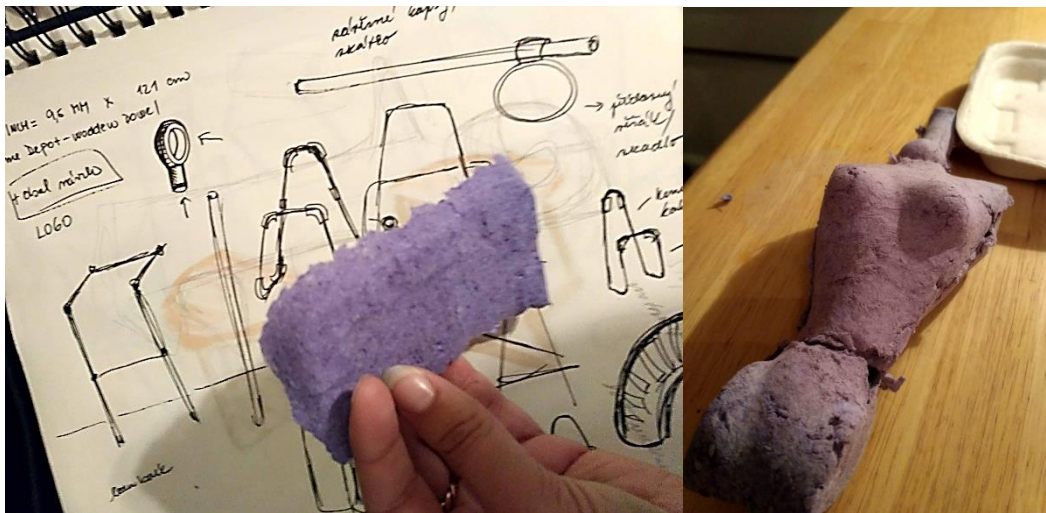


Figure 1 First experiments with paper pulp (author's archive, 2018)

After the arrival from the USA, the author had a one-year grant Internal Grant Agency (IGA) at Faculty of Forestry and Wood Technology (FFWT). The main goal was to reuse paper for wood-based products. The author went for a three-month internship to Salzburg University of Applied Sciences (SUAS), where a lot of interest is given to the topic of sustainability. The author had an opportunity to use the necessary equipment in the laboratory of SUAS in Kuchl, Austria. The research went into depth with the use of wastepaper in the engineered wood products area, where the paper could be used for creating MDF like boards which can be used for product design. The main goal was to manufacture paper boards and test its physical and mechanical properties. It was focused on the manufacturing and testing. The two types of pulp were used – old corrugated cardboards (OCC) and deinked pulp (DIP). Different adhesives were tested such as starch adhesive and urea formaldehyde glue (UF). Several boards were manufactured, and its physical and mechanical

properties were tested. The results of this research were presented at the international conference *"Wood Science and Engineering in the third millennium"* (ICWSE) 2019 in Brasov, Romania. An article *"Furniture design with principles of sustainable design by using paper pulp"* (Sobotková, et al., 2019) was also published in a special number of scientific magazines called ProLigno 2019. Vol. 15. nr. 4 in December 2019.

The project continued at Mendel University in Brno, where high energy milling (HEM) machine was used to mill the paper collected from the recycling centers around Brno, Czech Republic. The types of paper used were cardboard (CB) and office paper (OP). They were mixed with bioplastic (PLA) and limestone. Its mechanical and physical properties were also tested according to European standards. These boards had better results in comparison with the glued boards and they were used for further development. The abstract with the topic *"Paper and Plastic Waste Composite Boards for Furniture"* was published as an abstract in the proceedings of the convention *"Society of Wood Science and Technology International Convention"* in Portorož, Slovenia in 2020 (held online due to the Covid-19 pandemic). Other results were presented in a scientific article which was published in a scientific journal BioResources in an article *"Mechanical and Physical Properties of Boards Made from Recycled Paper"* (Sobotková, et al., 2021).

The research of the author was also influenced by the teaching experience of the author at the Department of Furniture, Furniture Design and Habitation, Faculty of Forestry and Wood Technology at Mendel University in Brno. The author is teaching 3d printing in English, graphic design, co-teaching Furniture Design and Interior Design subjects. The whole work was influenced by the 3d printing technology which corresponds with the topic of sustainable manufacturing process.

Another goal was to help with the organizing of international design workshops with foreign universities – first the one mentioned above – with LSU and second with SUAS. During the workshop with SUAS also very

successful Czech company TON from Bystřice pod Hostýnem participated. They provided waste materials from their production to the students to design new products from these materials. The results of the work were for example a festival chair, sunscreen, and vertical garden.

The author learned a lot of information thanks to these workshops especially since the topics were also focused on the sustainability and upcycling of salvaged materials. At Figure 2 there can be seen the salvaged parts of famous chairs no.14 from TON.



Figure 2 Salvaged part of chairs from iconic chair no.14 (AVC MENDELU, 2019)

The goal of the research was to create materials from waste which would be either burned or landfilled. The research started with the idea of using 3d printing technology along with waste materials, especially paper. Since this technology would be very hard to reach in the Czech Republic, the author has continued with the idea of reusing wastepaper by using another technology – pressing by hydraulic pressing machine. During the research, the author came to the development of composite paper-PLA boards. The use of the manufactured boards was supposed to be for wood-based products and that is why this thesis is mostly dealing with the important physical and mechanical properties of the boards and its application for furniture. The boards are not supposed to substitute industrially manufactured wood-based products. They are supposed to be another option for those who want to help

with the sustainable development, and it should show another way of thinking regarding to designing new interior products.

The idea was also to find companies which are interested in this topic and would like to participate as well. A company like that is Plastia s.r.o, which provided recycled materials from their products for further testing. Due to the circumstances, it was not possible to do more testing (pandemic situation in 2020). In the end the cooperation led to the common exhibition at Designblok 2020 in Prague where the whole exposition was done in cooperation with Plastia s.r.o. and the Department of Furniture, Furniture Design and Habitation. The whole exposition was focused on the topic of sustainability and circular economy. It was supposed to show the chain, where the paper is created, used and then it goes to the waste. The process does not end here though. The paper is then reused as a material for pressed boards together with plastic material (in this case it was PLA, biodegradable plastic). Later, these boards are used for furniture design. Here comes the company Plastia, which is also trying to use their waste materials. They are using recycled post-factory and post-consumer plastic for their new flowerpots. This cooperation has brought another idea of post-factory and post-consumer plastic HDPE (High Density Polyethylene) as a binder for the paper boards. The author went to the project "AKTION" to SUAS again to manufacture and test these boards in February 2020. Due to the pandemic, which came quite soon after the arrival, it was not possible to finish this research. After that the author continued with the tests at Mendel University in Brno where she tested if there are any VOCs (Volatile Organic Compounds) present in the boards. The results can be seen in the Chapter 5.

Finally, the boards were used for the design of collection of coffee tables PapeerStack which were exhibited at Designblok 2020, held from the 7th October till the 11th October 2020 in Prague and the author was able to present her work at the Mendel's University stand, which was the first time since MENDELU represented at this big design event. The author got very valuable notes from potential users/manufacturers to the project thanks to this event.

3. Literature review

In this chapter there is a description of the current state of problems with ecological materials, and it includes market research on the topic of bio-based and ecological composite materials. Several studies and research have been made about the topic of substitutes for wood-based materials.

All designers are thinking about the materials used for their designs. They are thinking about the materials properties and how they can be used. Some of them go even more into depth and research new options of the materials which goes in one hand with new technologies (Parsons, 2008). One of the most used materials for furniture is wood. Wood is a natural material, which is eco-friendly, but it is not a bottomless well. It can cause an ecological problem because of its disposal as well. The biggest issue with the wooden furniture comes during its manufacture. Therefore, its life cycle should be considered while using this material, which applies also while using any other material for the manufacturing of new products (Uysal, 2014). Many important terms regarding sustainability in general should be considered while designing new product. The important terms are Life cycle; Life cycle Assessment (LCA); Recycling; Downcycling; Upcycling; Cradle to cradle. Following pages are describing these terms.

3.1 Sustainability

According to the United Nations' World Commission on Environment and Development, sustainability is described as *"design that meets the need of the present generation without comprising the ability of future generations to meet their own needs"* as Uysal (2014) says in his research. Sustainability has three dimensions:

- **Economic**
- **Social**
- **Environmental** (Uysal, 2014)

The most discussed dimension in this thesis is environmental. Lately the companies and different organizations are trying to reduce their environmental impact which varies depending on the rules of each country.

Since the Industrial Revolution there have been many attempts to find more friendlier approach in the production. The pollution was so bad that the manufacturing had to be controlled not to cause sickness and death. Further development has led to a very important approach in design – eco-efficiency, which basically means *"do more with less"*. One of the pathfinders was Henry Ford with his clean operating policies and reducing waste (Braungart, et al., 2002). There has been a huge boom with biodegradable materials and many materials are being recycled. The problem with biodegradable materials is, that often they are releasing toxins into the environment and actually, most of the recycled materials are downcycled. It means that the quality of the material is reduced over the time. The other disadvantage is that there is increased the contamination of the biosphere (Braungart, et al., 2002).

Every material has its life cycle which starts with their creation and then is used for manufacturing of a product. These products, like humans will also come to an end of life. Although the materials from which they are made are going to stay, some of them can be reused and other products can be made (Ashby, 2009).

3.2 Circular economy

According to the *"Institute of Circular Economy"* (INCIEN) (Jonášová, 2021), circular economy is a sustainable development strategy, which creates a functional relationship between nature and human. The main idea is to create circular flow between the materials, products, and the end of its lifetime. In a current linear system, the problem is, that the materials are used for the manufacturing of products, which are sold, and they are either burned or they end-up in a landfill after its lifetime ends. The reason why circular economy is growing also in the Czech Republic is because we are realizing that natural resources are valuable, and we should not waste them. Designers should think about other options for their customers or users and what else the products can serve for (Bakker et al., 2019).

There are many companies which are already using the main rules of circular economy in their production. One of the examples in furniture industry is one of the biggest furniture sellers globally – IKEA. The company started to focus on a circular economy, because they do not want to contribute to the climate change anymore and that is why circular business is influencing their business in all aspects. Their goal is to offer 100 % circular products by the year 2030. They want to use either recycled materials, renewable or return the products, when the customers do not want them anymore (Jonášová, 2021).



Figure 3 Circular economy (Jonášová, 2021)

7 rules of circular economy

There are seven key rules of circular economy. Each one of them is described below:

- 1) Change of business model
- 2) Extend life cycle of the product
- 3) Eco-design
- 4) Use of renewable resources
- 5) Cooperation with suppliers and customers
- 6) Use of digital technologies
- 7) Waste as a source

3.3 Eco-design

Eco-design was motivated by the increase of awareness about the environment and its worsening state. From several analysis during the 80's it was discovered that the most efficient and economic are the protections, which are based on the prevention, which means to look for the reasons and eliminate the reasons of its pollution. Because of that, the manufacturing process has become cleaner, and environmental policies were developed. It can be explained as a „*systematic integration of environmental considerations into product and process design*“ as Costa, et al. (2003) refers in their research.

The main idea of eco-design is presented by Figure 4, the eco-design strategy circle of the products life cycle. The eco-design strategy circle can provide a roadmap for the creation of composite materials from recycled paper materials. Eco-design strategies can help the businesses to reduce the environmental impact, optimize material use and energy use, optimize the distribution and production, reduce production costs and so on (Costa & Gouvinhas, 2000).

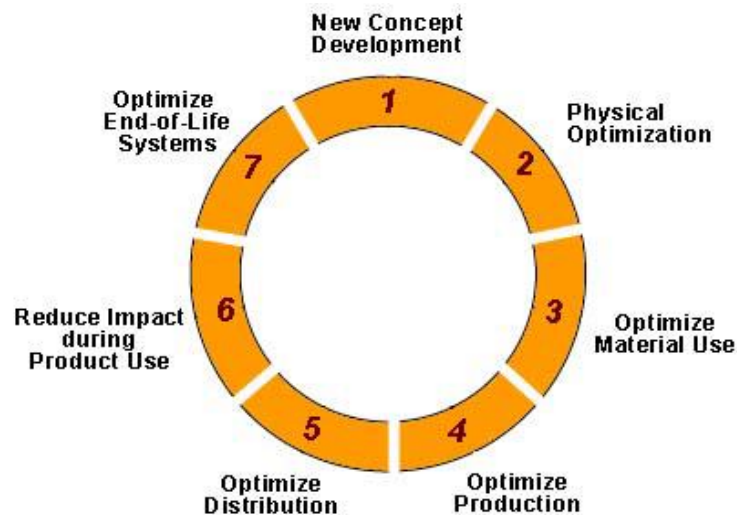


Figure 4 Eco-design strategy wheel (Costa & Gouvinhas, 2000)

3.4 Lifecycle

The lifecycle of every material and product is like the human life. They are born, they develop over time and grow, they get old, and they die in the end. Every product is made from materials. The lifecycle of each material is tracing its development and end of lifetime, although not all materials have to be disposed at the end of their lifetime and they can be reused for other purposes (Ashby, 2009). At the Figure 5 there can be seen the lifecycle of a product, which starts with the materials, productions, transport which leads to the use of a product and then comes the end of Life. The material lifecycle shown at

Figure 6 is describing the process of a material's lifecycle even more. At the beginning there are the natural resources and feedstock, which are then manufactured into a product. They are being used until the end of its lifetime, and then they are discarded or recycled. In every phase of the lifecycle, there is energy consumed and other waste is generated together with emission such as carbon dioxide (CO_2), oxides of sulfur (SO_x), Nitrogen (NO_x), and other emissions. They are non-harmful in a small amount, but the more materials and products are produced the more the emissions are as well (Ashby, 2009).



Figure 5 Lifecycle of a product (Ashby, 2009)

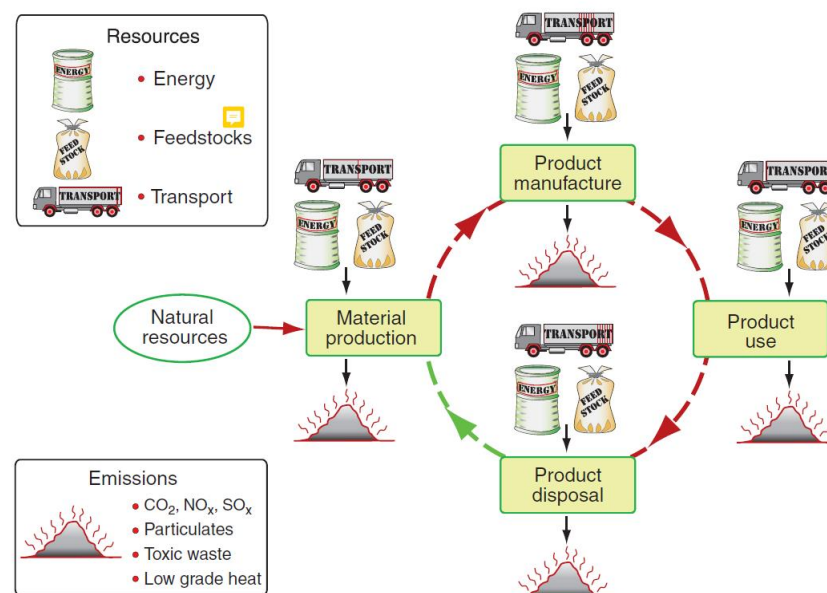


Figure 6 The material lifecycle (Ashby, 2009)

An example of a carbon footprint of different materials can be seen at Figure 7. According to Ashby (2009), the big four materials which are emitting the biggest amount of CO₂ are iron and steel, aluminum, concrete (cement) but also paper and cardboard. In contrast with this there is the recycling chart at Figure 8, which is showing that the recycling of metals and paper is very developed. On the other hand, the recycling of polymers, concrete, but also

wood is about 5 %, although it has changed in the last years, but there is still a lot of plastic waste which is not recycled.

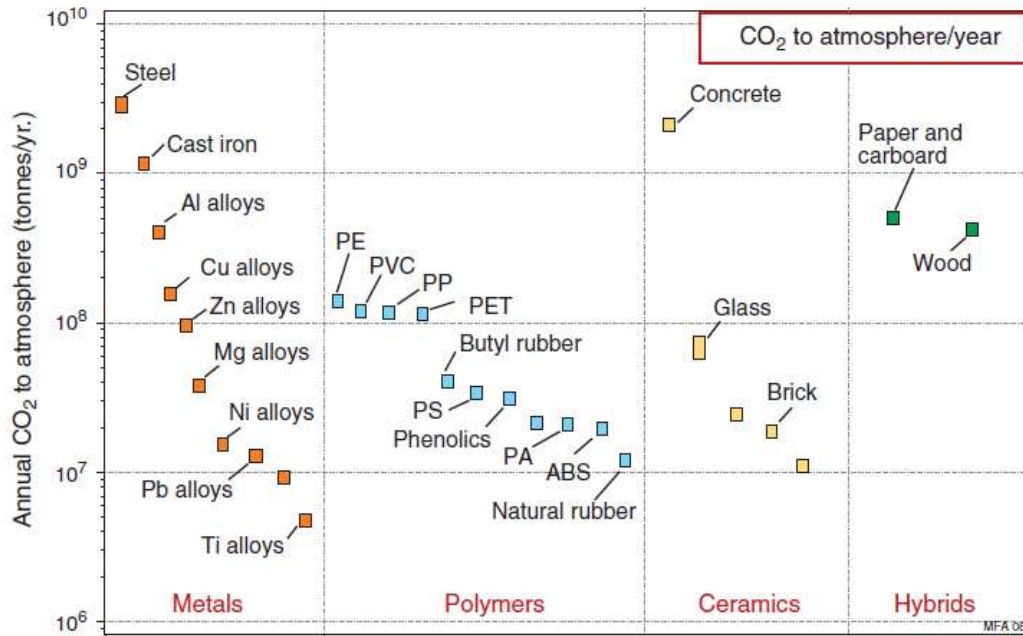


Figure 7 Annual carbon dioxide to atmosphere from material production (Ashby, 2009)

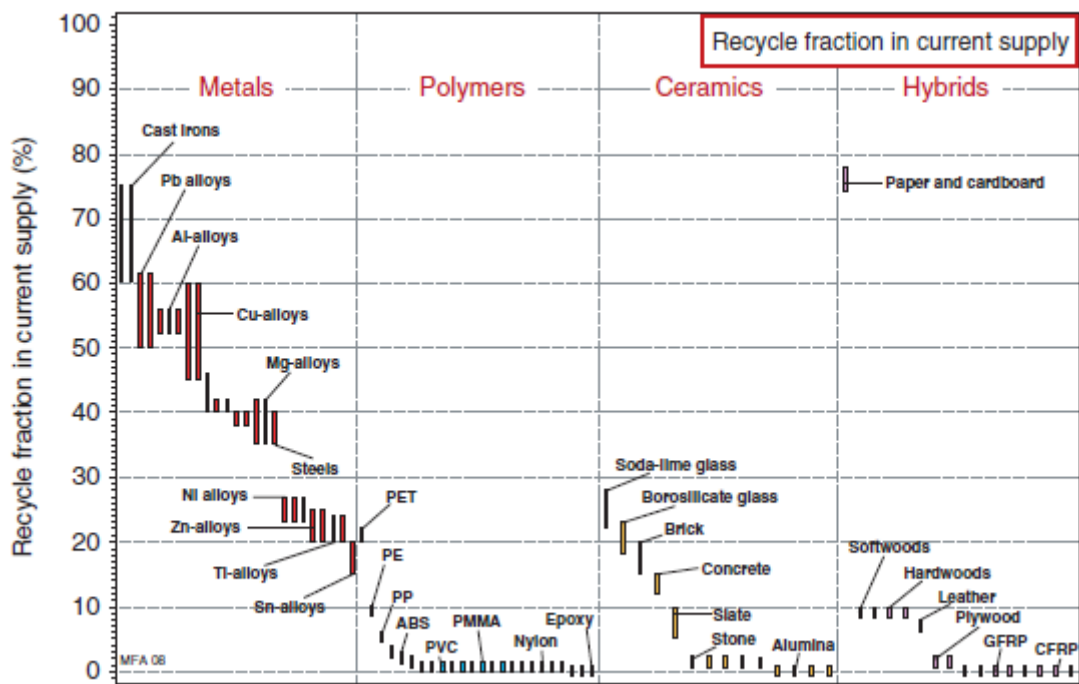


Figure 8 Recycle fraction of materials (Ashby, 2009)

3.5 Life cycle assessment (LCA)

Life cycle assessment (LCA) is a tool which can document consumed resources, emitted emissions and their impact during each phase of a materials life cycle. LCA is a tool which can help with the choice of material during the creation of a new product. Basically, it can answer the question *"How sustainable is my product or process?"* LCA helps to determine the environmental impact of the material, product, or process (One Click LCA, 2020). At Figure 9 there is visualized the process of LCA, which starts with the resources. After that, the processing part comes, and it is followed by manufacturing of products. When the product is ready, the distribution part comes and then it can be used by the customers until its end of life. Here comes the important part of having the other purpose for the product or material.

To determine LCA of any material or product there are so called eco-indicators which are used during the process. To determine the products or materials LCA there are many types of software which can be used like SimaPro (2008) but also an online tool.

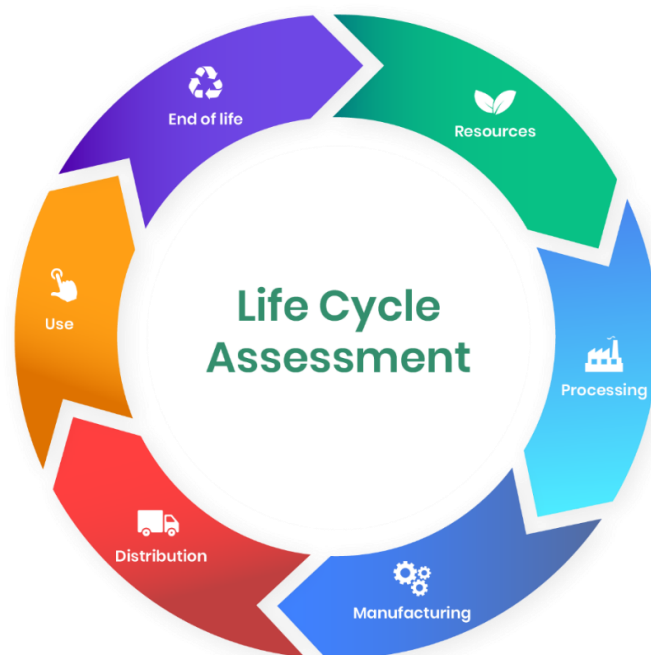


Figure 9 Life cycle assessment stages (One Click LCA, 2020)

3.6 End-of-life strategies

According to Uysal (2014) there are many questions which we should ask after the end of the life cycle of every material or product. Ideally the end-of life strategies should be used while designing a new product and the products should not end up in a landfill. The problem is that in many countries the landfill is full already.

Several important questions which should be asked after the End-of-life of each product are described below:

- Can the product be reused somewhere else?
- Can the damaged product be fixed with new parts?
- Can the damaged parts be used for the fixing of another product?
- Can salvaged materials be used for other products or for the creation of other materials?

The companies should also follow the important questions for sustainable design (Bras, 1997), namely:

- Simple – easy to use.
- Easy obtainable – affordable and reasonable cost
- Precisely definable – obvious how to use.
- Objective – same results should be obtained.
- Valid – accurately measured, indicate, and predicts what is intended.
- Robust – insensitive to changes in the domain of application.
- Enhance understanding and prediction.

The concept of End-of-life aims avoid or minimize impacts to the environment (Uysal, 2014).

3.7 The four's R

There are several approaches to the environmental designing, e.g., recycling, biodegradable materials, or low energy consumption according to Šimek (2009). Four very important rules in a sustainable design process and a way of living are more and more discussed. The rules are to a) reduce, b) reuse, c) recycle, d) regulate. Sorting out the waste and its recycling is done on a regular basis. The most often recycled materials are paper, plastic and glass in the Czech Republic. In other countries such as Austria, Finland and other European countries also sorting out the biological waste and metal is done on regular basis.

3.7.1 Reduce

Reducing the number of used products and materials is important for our polluted planet. One example can be reducing dangerous emissions released from particleboards. The emissions have been reduced to a minimum of the emitted formaldehyde from the boards since after many studies have been made and it was found that formaldehyde and other Volatile Organic Compounds (VOCs) are carcinogenic.

3.7.2 Reuse

Reusing materials and products is not a new idea and it has been here even during the ancient times. Old structures were used for building new ones. The owner must be willing to be satisfied with its „*already used state*“. We can see this with used cars, which are also cheaper than new ones, secondhand clothes and furniture. Even though it was used already by someone else, it is still possible to refurbish it or upgrade it, so it looks good again. In the case of reusing, the state of mind of the user must be changed so that they believe that even though the product is old, it can still be fixed or reused and can have another purpose. Unfortunately, we live in a materialistic time, when it is too easy to throw away things that are already broken and buy a new one rather than to fix them.

Lately, many people and companies have started realizing these problems but also designers have started to focus on reusing old products or materials.

3.7.3 Recycle

In developed countries, recycling became a natural part of their lifestyle. There are waste containers at almost every corner. Nevertheless, most of the recycling is downcycling because the quality of materials is reduced. Unfortunately, even if the material was recycled, this does not mean that it is ecological. But many materials can still be reused even if people do not realize it yet.

Metals. The recycling of metals is highly developed, and their recycling is economically attractive. The reason why they are so easy to recycle is because of their density, magnetic and electrical properties and even color differs a lot from others. There are limitations to the use of metals after they are recycled but there are enough good uses so their contribution to recycling and today's consumption is large (Ashby, 2009). The problem comes with the paint which is melted together with recycled metals. It consists of chemicals which are harmful to the biosphere.

In the Czech Republic, the metal packages such as cans from food or beverages can also be sorted into the yellow containers. In comparison with Louisiana, USA where the author lived, there was only one regular trash can for sorting all plastics together – metals and glass.

Glass. Glass is a material which also can be recycled. In the Czech Republic there are green containers for sorting glass waste –there is also a white container for white clear glass and a green one for colored glass.

Polymers. The same cannot be said for polymers. The manufacturing of polymers or in other words plastic materials is huge. Many of the manufactured materials have a very short life cycle and they are a major issue in waste management. Unlike metals, polymers all have nearly the same density, they do not have any significant magnetic or electrical signature and they can be any color. Its recycling is involving a big amount of energy

consumption (Ashby, 2009). In the Czech Republic there are yellow containers available for sorting out the waste plastic materials. It is recommended by each city, town, or a village how to sort out the waste there.

Paper. Even the recycling of paper is a problem because it was not designed with recycling in mind. Recycling of paper requires bleaching and other chemical additives for making it useful as a paper again. The result is a mixture of pulp, chemicals and other additives which are making the recycled paper harmful (Braungart, et al., 2002).

In the Czech Republic, there are blue containers to throw paper away. The paper which can be sorted this way is newspaper, magazines, flyers, cardboard boxes, cardboard, office paper, notebooks and so on. The paper cannot be dirty or greasy. In comparison with Austria, most containers have the same color, except paper – the container there is red. The greater the amount of people who are consuming the products and materials, the greater the waste. But the waste should be another resource. Even though the products life has come to an end, there are still parts of materials which are useful. The problem is that even though the materials are recycled, they are sometimes mixed and burned.

In the Table 1 there can be seen the recycling markets according to (Ashby, 2009). There are the developer end uses for recycled materials for metals such as steel, cast iron, aluminum, copper and so on. Also, polymers such as Polyethylene terephthalate (PET), High-density polyethylene (HDPE), Polypropylene (PP) and Polyvinylchloride (PVC) are recyclable. There are also some existing secondary uses, but they are not developed as a market. For example, the use of textiles or paper-metal foil packages.

Table 1 Recycling markets (Ashby, 2009)

MATERIAL FAMILY	DEVELOPED END USES FOR RECYCLED MATERIALS	EXISTING SECONDARY USES BUT NOT DEVELOPED AS A MARKET
METALS	Steel and cast iron Aluminum Copper Lead Titanium All precious metals	Paper –metal foil packaging
POLYMERS AND ELASTOMERS	Polyethylene terephthalate (PET) High-density polyethylene (HDPE) Polypropylene (PP) Polyvinylchloride (PVC)	All other polymers and elastomers, notably tires
CERAMICS AND GLASSES	Bottle glass Brick Concrete and asphalt	Nonbottled glass
OTHER MATERIALS	Cardboard, Paper, Newsprint	Wood Textiles: cotton, wool, and other fibers

3.7.4 Regulate

Another important rule is to regulate the use of materials and products. We live in a materialistic time with many tempting options and the advertisements are also doing their job to get more and more customers. People are buying products that they do not need, which causes environmental problems. The best option would be to support local manufacturers which are using quality materials for their production. These products can have a longer lifetime and people would not need to buy new things all the time. That would help to regulate the amount of the waste products and materials too.

3.8 Downcycling

Downcycling can actually be more expensive for business, because it is trying to give materials a longer lifespan than they were designed for. Eventually they are incinerated or landfilled anyway. In an ideal world the products would be made with thought of upcycling them after their typical use. One example of this idea is the creation of waterproof paper by Charlie Melcher of Melcher Media. Paper which is derived from polymers is waterproof and printing inks are non-toxic and can be washed out completely with water after

its use. This type of paper can be upcycled later with the same value and properties (Braungart, et al., 2002).

3.8.1 Cradle to grave

Most of the products are created in the cradle-to-grave model. The products itself were made from valuable materials; they have created valuable products which are supposed to earn money but in the end of their lifetime they end up in a landfill with no other use. That means cradle to grave. Even though, for example, food and paper can decompose and are heaped in a landfill or incinerated and they no longer bring any value (Braungart, et al., 2002).

3.9 Upcycling

Upcycling is a new way of thinking about why we should design new products. The main idea of this process is to transfer old products into new ones with higher use quality than the previous product's quality. An example can be the approach of young design studio NAHAKU from Brno, the Czech Republic, which started their work with idea of using old extinguishers for bowls and other decorative products. They continue to develop very interesting products made of salvaged materials. Czech companies are trying to design new products in a sustainable way from quality materials. An example can be TON company, which is known for their quality chairs from bent beech wood. They have provided broken weaved seats from their chairs to the company NAHAKU which designed light for interior of a wedding venue (Figure 10). At Figure 11 can be seen their design of lighting made of old buckets for prosecco.

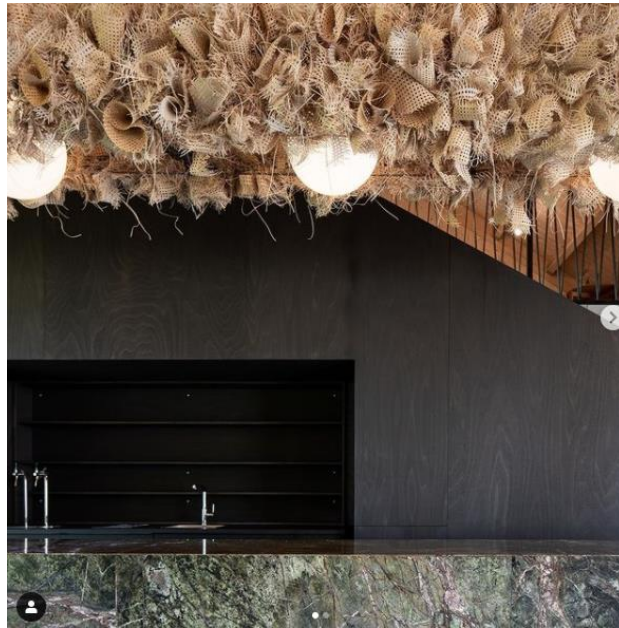


Figure 10 Reused weaved seats for TON chairs (Nahaku, 2020)

An example can also be the company Preciosa, which is very famous for their manufacturing of jewelry. They provide their small beads waste to the Czech designer Sára Matysová, who designed a beautiful mirror stand out of this waste material. It is great to see that companies are thinking about these problems and helping young designers to bring their ideas to life.

3.9.1 Cradle to cradle (C2C)

C2C is an important approach to sustainable development. The main idea is to design objects which will have fulfilling life cycle, but after the end of their lifetime the materials could be still used for creating another product. It is the opposite to the approach cradle to grave, which is a symbol of most developed products on the market. Even though companies are often manufacturing products which are serving for a limited amount of time and then they are simply thrown away.



Figure 11 PETing - Lights made from old buckets for prosecco (NAHAKU, 2021)

3.10 Environment and sustainability

In the last years, the environmental concerns have grown, and the need of reusing waste materials brought many so called “*eco-friendly*” or recycled materials (Tlili, et al., 2014). The environment where the people live should be ecological – according to Caplan (2016) it should benefit human welfare with minimal impact to the natural environment.

One of the examples is Wood Plastic Composite (WPC), which is very similar to the materials tested during this research. Together with the question of new eco-friendly materials there comes a question if they are non-harmful to the environment. During the creation of every new product like the materials mentioned on the previous pages it is necessary to think about its ecological impact also on our environment.

3.10.1 Volatile Organic Compounds (VOCs)

There are many factors which can affect the environment. One of them are VOCs and Total Volatile Organic Compounds (TVOCs) (Čech, 2008). VOCs are loosened up into the air during the manufacturing process but also during the use of new furniture. These compounds play quite a role on the pollution of our environment. There are many chemicals in the air, which are from the

incineration, gasoline, power plants and others. Also, on any interior there are many chemicals on the materials within. One of the most disputable was and still is formaldehyde, which is carcinogenic (Čech, 2008). According to The World Health Organization (WHO) (1989), VOCs are characterized as compounds with a boiling point between 50°C to 260°C. VOCs are for example benzene, toluene, xylene, chlorinated solvents and so on. The separate group is formaldehyde because of its impact on human health (Čech, 2008). Formaldehyde has been one of the greatest problems because of its bad influence on human health for many years, even though it has not been very clear how much. It is considered as potentially carcinogenic substance. There has been a lot of research made and there are many restrictions now on the use of glues such as urea-formaldehyde glue in the furniture industry and wood-based panels, so the emissions are the smallest. The best way is to reduce the emissions of chemicals and combine other factors which create the interior environment (Brunecký, 1998).

Another term is TVOC, which means the total amount of Volatile Organic Compounds. According to Tesařová, et al. (2007), TVOC value indicates the level of indoor pollution and it can be measured by the gas chromatograph.

According to Čech (2008) the greatest focus is given to the VOC in interior, especially in case of constructional materials, surface finishes and also the surfaces which come into the contact with food. Since the paper-based boards that were manufactured and tested during the research are designed for the interior use and they can also come in touch with food, it was important to know, how many VOCs might be emitted from the boards.

During the research the testing of VOCs for the manufactured paper-PLA boards was done. Some of the chosen VOCs are mentioned below:

- a) Toluene C_7H_8 – the steam of toluene has narcotic influence and they cause a headache, stomach ache and respiratory sickness.
- b) Xylenes C_8H_{10} – organic solvent, can be found in the surface finishes.
- c) Butyl Acetate $C_6H_{12}O_2$ – organic compound, ester.

- d) Ethyl acetate – organic compound which can be found in pears.
- e) Pentanal $C_5H_{10}O$ – organic compound, aroma.
- f) Hexanal $C_6H_{12}O$ – aldehyde, can be found in the wood.
- g) Limonene – natural substance, found in the needles of the trees, often used as aroma in the perfumes, detergents and so on.

3.11 Material research

During the research it was necessary to find more information about materials which can be used in a sustainable way. Designers of the 21st century should try to minimize the ecological burden (Ashby et al., 2002). That is why the materials and market research was done. Many researchers are focusing on sustainability in their research. There have been several scientific articles made about the recycled materials for ecological boards which could replace regularly used boards in the furniture industry. The need of sustainable products is non-disputable. On the following pages there will be a description of several different materials used by researchers and companies, which are already using them for their products. There are materials mentioned which are industrially manufactured and being sold under their trade name. Many companies have noticed the need of environmentally friendly products but also the materials and how they are focusing together with scientists on the development of substitute materials which will not harm our environment anymore. Another part of the research was to find products, mostly furniture which was made from paper or paper pulp. The biggest inspiration for this work was a Canadian company Dear Human, which is creating furniture or wall coverings such as tiles out of paper pulp.

3.11.1 Rice straw

There are many biological materials – waste residues of plants, which can be a very good source for production of bio-based boards in the engineered wood products industry. Materials such as rice straw, husk, bamboo or banana leaves were being used for creating such products. According to Fernandez-

Garcia, et al. (2017) who created rice straw particleboards, its physical and mechanical properties were tested in a similar way as in the case of regular wood-based products. During their research, binderless boards were manufactured through hot pressing at a low temperature of 110°C and pressure of 2.6 N/mm². The boards could be used as thermal insulating panels (Fernandez-Garcia, et al., 2017).

3.11.2 Resysta

Resysta is a bio-based material made of 60 % rice husk, 22 % common salt and 18 % mineral oil. It is a durable material which is used by a Czech company mmcité for highly strained outdoor furniture. Its visual look is wood-like although it requires minimal maintenance, and it is also highly resistant to pests, mold, or cracks. The company mmcité is using this material for the design of their benches, bins, and planters (mmcité, 2020).

This material was developed by the company Resysta, which is substituted by the company Limany s.r.o. in the Czech Republic. Their aim was to develop materials which will have a similar look as the tropical wood, since rain forests are constantly being cut down, but they are important for the whole earth.

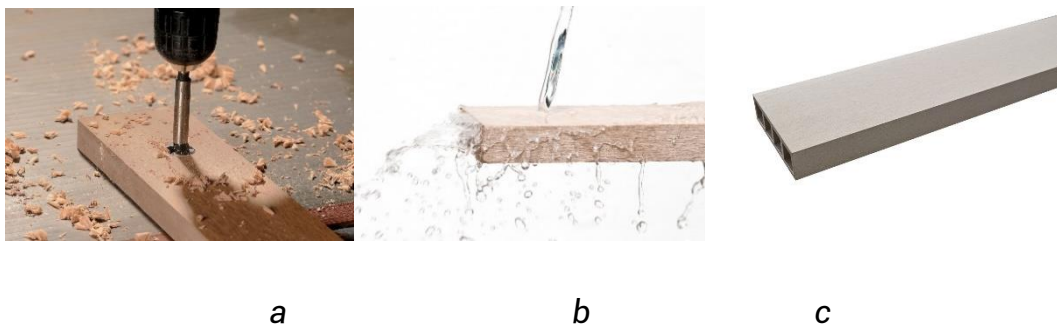


Figure 12 Resysta; a -tools; b - water resistance; c – profiles (mmcité, 2020)



Figure 13 Bench for parks Vera, Spain (mmcite, 2020)

3.11.3 Wood plastic composites (WPC)

According to Shalbafan et al. (2016) using waste production has economic and ecological importance in the particleboards industry. One of the examples are wood plastic composites (WPC). They were first manufactured in the 1990's as a homogeneous material from a mixture of wood particles and thermoplastic polymers, such as polyethylene (PE), polypropylene (PP), polyvinylchlorid (PVC) or polylactiacid (PLA). Composites made out of these materials are durable and potentially recyclable. WPC with the right amount of adhesive can be also used for outdoor use (Shalbafan, et al., 2016). WPC boards are regularly used for terraces decking, stairs or fences, but they can also be used indoors.

3.11.4 Recycled plastic

Urban chic stools (Figure 14) are made from 100% recycled plastic waste materials by Chilean designer Rodrigo Alonso. The stools are made from old drink trays, stadium seats or even electronic devices. These waste materials were then transformed through a molding technique into useful objects such as eco-modern stools or recycling bins. At the end of its lifetime, the stools can be re-crushed and re-molded again into another functional piece.



Figure 14 Urban-chic stools (eco-chic-design.com, 2020)

3.11.5 Algae

Dutch designers Eric Klarenbeek and Maartje Dros have developed a bioplastic material made from seaweed algae. They believe that this material could replace synthetically manufactured plastic materials. After a long process of research, they developed a 3d filament from algae which can be used for 3d printing of biodegradable pots, shampoo bottles and so on. They believe that this type of biobased plastic could replace fossil fuel-based plastic. An example of using this material can be seen at Figure 15.

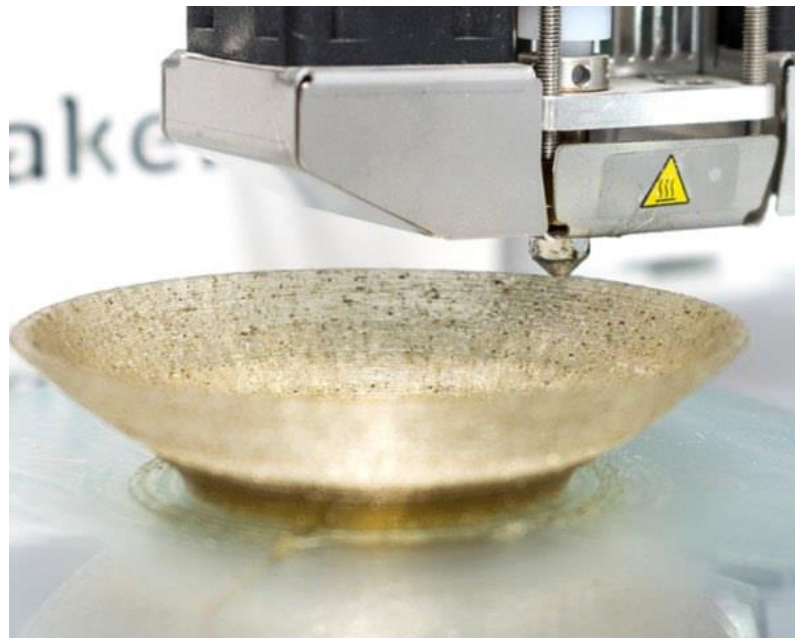


Figure 15 Algae 3d filament (dezeen.com, 2020)

3.11.6 Mycelium

Mycelium is a type of mushrooms which can grow into a product, even furniture. According to Abhijith (2017) it is possible to manufacture packages from mycelium, natural glue which latches onto particles around it. These packages can serve as a substitute for polystyrene or other plastic foams where the main component is derived from petroleum or natural gas. The mushroom roots can grow in a mold and form into different shapes. This type of material is very dense and strong. After it has grown into the desired shape it is dehydrated to stop the growth of the mushroom (Abhijith, et al., 2017). The same material was used by the Czech studio LLEV (Figure 16), who exhibited their products – mostly bowls, but also small coffee tables made of mycelium at Designblok 2020 in Prague. The exposition was designed together with an Austrian company Organoid, which is developing boards for furniture and interiors from dried weed or meadow flowers. A Czech company Dřevojas also used these boards for their bathroom furniture. Another great example of the use of mycelium is a project of a polish student Agata Szlag, which took a class of Furniture Design taught by the author of the thesis. She has experiment with mycelium mushroom and used also FDM 3d printing machine to print a form for growing the mushrooms. The process can be seen at Figure 17.



Figure 16 Coffee tables by LLEV studio at Designblok 2020 (author's archive, 2020)



Figure 17 Project of Agata Szelag from mycelium (Agata Szelag, MENDELU 2019)

3.11.6.1 *MycoComposite™*

Mushroom based materials made from mycelium are biological materials used as a binder of other materials. It was developed in 2007 by Ecovative design. It is used as a sustainable packaging material 100 % biobased and C2C Gold certified. It only takes 9 days to grow, and it can grow in a custom-made mold. These materials are fire resistant, biodegradable, and hydrophobic. MycoComposite boards have the potential of replacing wood based products where mycelium would be used instead of resins (Abhijith, et al., 2017).



Figure 18 MycoComposite™ (Ecovative, 2020)

3.11.7 Paper

On the following pages there will be described different materials based on paper. There is the description of the paper pulp, which played a big role in the whole research. Also, many companies are using paper pulp or paper-based materials in their design processes.

The forerunner of paper is papyrus, which is made from the flower stem of the reed. It has been known and used for over 5000 years, native to Egypt. On the other hand, paper is a Chinese invention (Ashby, 2009). Paper is a material which is manufactured worldwide. It is a cellulose-based material, it starts from sheets of office paper, packages, newspaper but also photography, money, and stamps. The main component of paper are fibers derived from plants where the main source is wood. The secondary fiber is derived from wastepaper and the recycled paper is made (Kirwan, 2013). Paper is formed from paper pulp fibers which are drained through a filter screen to form a fibrous sheet. From the filtering mesh the paper sheet is then pressed through a cylindrical roller for higher density and squeeze out the water (Xia, 2002). In the furniture industry paper is used as impregnated paper for the top layers of wood-based panels. Packaging material is mainly made from paper where up to 40 % is made from paper or paperboard. Paper packages can be found as the packaging for dry food products (cereals, tea, coffee etc.), frozen food such as ice cream, liquid food such as milk, fast food packaging, personal care, sport, and leisure equipment and more (Kirwan, 2013). Some general properties of paper can be seen below (see Table 2, Table 3, Table 4, Table 5).

Table 2 General properties of paper

Density	480–860 kg/m³
Price	2.07 – 12.4 USD/kg

Table 3 Mechanical properties of paper

Young's modulus	3–8.9 GPa
Yield strength (elastic limit)	15–34 MPa
Tensile strength	23–51 MPa
Compressive strength	41–55 MPa
Elongation	0.75–2 %
Hardness –Vickers	4–9 HV

Table 4 Ecoproperties: paper material

Annual world production	3.6 x 10⁸–3.7 x 10⁸ tonne/yr
Embodied energy, primary production	24.2–32 MJ/kg
CO₂ footprint, primary production	1.23 – 1.55 kg/kg
Water usage	1100–1200 l/kg
Eco-indicator	91–100 millipoints/kg

Table 5 Recycling of paper

Embodied energy, recycling	18–20 MJ/kg
CO₂ footprint, recycling	0.72–0.78 kg/kg
Recycle fraction in current supply	70–74 %

Typical uses. Packaging, filtering, writing; printing; currency; electrical and thermal insulation

3.11.7.1 Paper pulp

Paper pulp is a cellulose-based material, and it is created from several manufacturing processes. The biggest amount of pulp (about 90 %) is made from wood and the rest from other plants. The pulping industry can also be seen as a great example of a self-sustaining industry with sustainable development (Sixta, 2006).

Cellulose fibers are bonded together by lignin from trees and complex polymer which combined make up to 30 % of trees. Paper pulp is derived from washed wood chips during the pulping process. The wood chips are then refined in the refiner and cleaned which results into final pulp (Figure 19). The pulping process can be either mechanical or chemical.

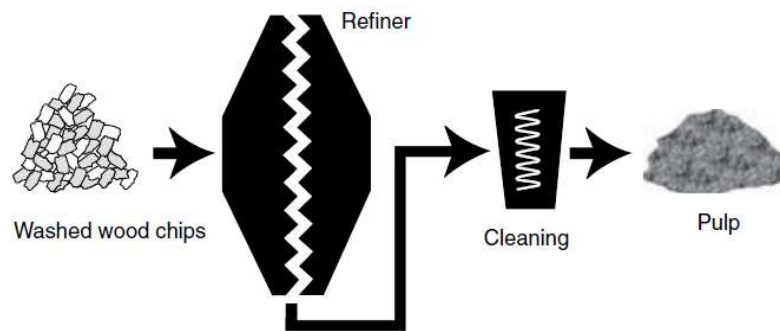


Figure 19 Production of mechanically separated pulp (Kirwan, 2013)

Paper pulp is one of the materials, which can be used as a sustainable source. Its applications in cellulose-based composites are countless according to Madyira (2017). In his research he focused on using paper pulp together with polyvinylacetate (PVAc) glue. He measured mechanical and physical properties of this composite material and found out that paper pulp and wood glue could be used for creating natural composite material. The strength of the material depends on the paper-pulp fraction and it could be used for structural applications subjected to low loading conditions (Madyira, et al., 2017). Paper pulp can be seen at Figure 20.



Figure 20 Paper pulp (author's archive, 2020)

3.11.7.2 Paper Sludge

The pulp and paper industry is producing high amounts of paper sludge which is produced during wood pulping and papermaking. Paper sludge consists of very fine fibres and some inorganic materials (kaolin clay, calcium carbonate) shown in Figure 21. At the picture A are the paper fibers and B are the minerals. This material can serve as a filler for thermoplastic polymer composites according to various studies. According to Taramian (2007), paper sludge lowered the mechanical properties of particleboards. It could be because of the inorganic materials in the sludge. This topic needs further investigation but it could also be a way to reuse waste material (Taramian, et al., 2007).

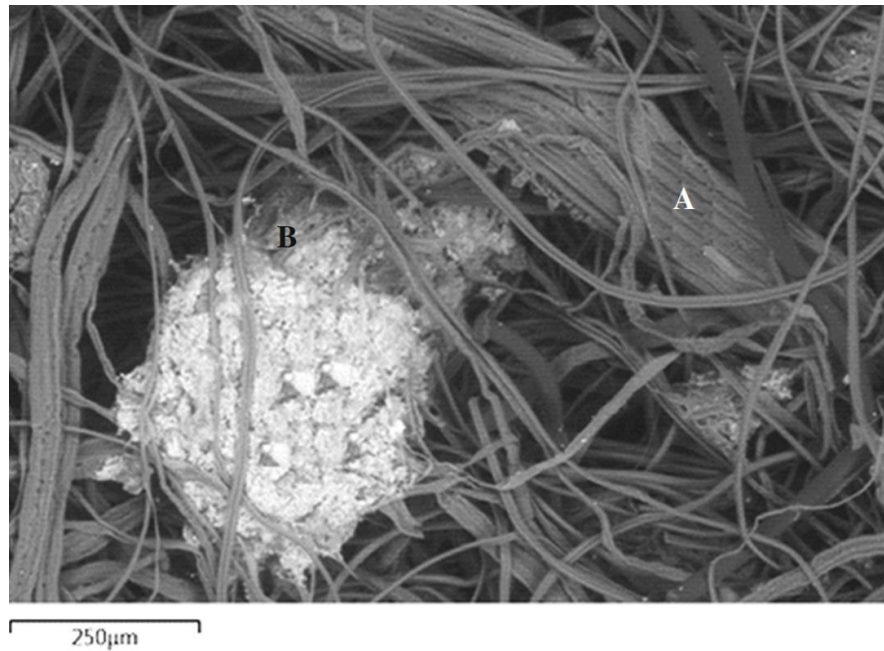


Figure 21 Paper sludge micrograph (Sustainable Design and Manufacturing, 2017)

3.11.7.3 Paper and Dear Human

The work of this Canadian studio is focused on the exploration through unconventional material use, pattern, and color. The studio was formed by Jasna Sokolovic and Noel O'Connell in 2010. It is a visual arts and design studio exhibiting their work in the famous design weeks across the world – New York, Milan, Stockholm, London, Tokyo, Montreal, and other places. They are using locally gathered paper from business offices in Montreal, Canada and producing paper tiles and paper pulp furniture (Figure 22a).

Paper tiles (Figure 23) are made of 100 % recycled materials made of collected post-consumer paper from cooperating local businesses. It is sound absorbent, and they are like acoustic foams which are available on the market. All tiles are made by hand. Tiles are as hard as board, light as cork and can be printed and painted on like paper. Instead of their decorative potential they are also absorbing sound and easy to install.

They also developed material called Paper-rock (Figure 22b). Paper-rock has been used for creating the stone-like surfaces of a table. They are experimenting with paper pulp furniture and creating acoustic wallpaper tiles.

Paper-rock consists of paper pulp and hardening agents. This material is very resistant and can be used for furniture. One of their projects is a coffee table with the top board being made from paper-rock. Their first attempt of reusing paper were paper tiles, which are shaped into panels which work as an acoustic barrier. The main source for the company is a wastepaper which they collected from local offices. Their design goes in one hand with sustainability and circularity.



Figure 22 – a - Paper pulp tables, Dear Human; b- Dear Human Paper Rock table (Dear Human, 2020)

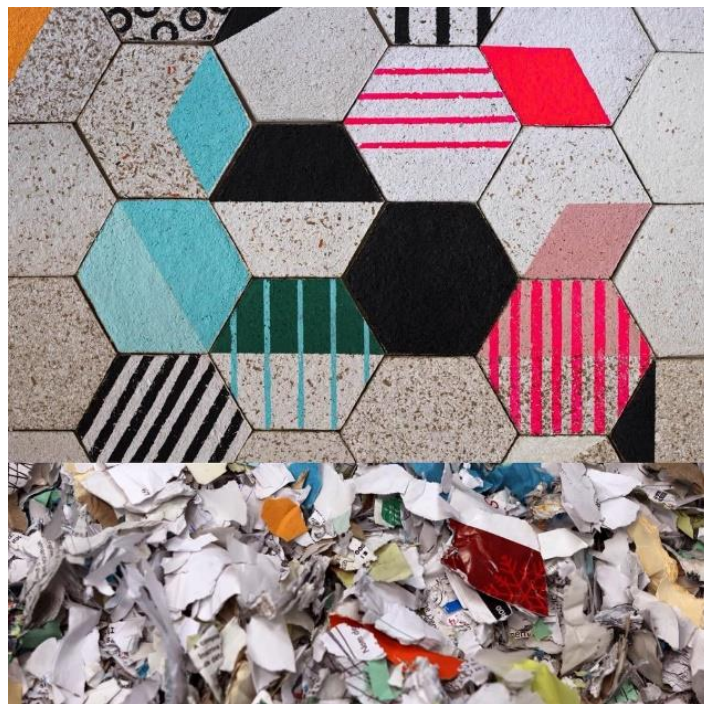


Figure 23 Paper wall tiles (Dear Human, 2020)

3.11.7.4 PINWU Design studio

PINWU studio is based in China, and it focuses on the deconstruction of traditionally used materials in China. The studio has created a chair (Figure 24) from glued layers of paper which come from traditional manufactured paper umbrella. It is made from handmade rice paper. The seat keeps its flexibility, but it is strong enough for sitting.



Figure 24 - PINWU paper chair (John, 2019)

3.11.7.5 Morgan Doty Design

Morgan Doty is a Melbourne based designer who graduated with a degree in architecture. As a part of her research, she developed an armchair and stools from paper pulp and wood (Figure 25). Her first success was a CMYK stool project. The project is exploring reusing of wastepaper for furniture product. Pulp is shredded and molded over CNC cut form. It is combining digital fabrication to gain extremely strong shell structure as a stool seat. The seat of the stool is easy to disassemble and can be replaced with another one. The surface is telling the story of used paper. As a binder there was used a biodegradable glue and for the construction under the seat was used American oak. The stools were exhibited at Salone Satellite 2019 at Salone del Mobile Milano 2019.



Figure 25 CMYK stool, Morgan Doty Design (Doty, 2019)

3.11.7.6 Paptic

Paptic is a new alternative to plastic materials in packaging. It is a material made of paper and plastic which the name already indicates. This material looks like paper, but it has similar properties as plastic and its very resistant. It is a bio-based material, which can be recycled, and it is also biodegradable.

This concept is built around six core aspects – renewable raw materials, reusability, recyclability, resource efficiency, biodegradability and last, not the least circular economy. At Figure 26 there can be seen bags made of Paptic material.



Figure 26 Bag from Paptic (Paptic Material, 2019)

3.11.7.7 ReWall

Another material with the trade name ReWall (Figure 27) is a composite material which is manufactured without any formaldehyde glue or other chemicals. It is made from 100 % post-consumer beverage packages. Since the cartons for milk or juices are made from paper, which is covered with aluminum foil, it is very difficult to separate them after the end of its lifetime. For this new composite material, the beverage cartons do not have to be separated. The materials are resistant towards mold and water. 300,000 beverage cartons are used to create one board.



Figure 27 ReWall (Healthy Materials Lab, 2020)

4. Methodology

In this part of the thesis, there will be described the used methodology of paper-based boards for manufacturing with the circular economy rules (3.2) and in a sustainable way with similar aim like in case of the researched materials. The author of the thesis wanted to show other options how to work with waste materials as described in the Literature review. The boards were created with the idea of sustainable development of material which could be used for furniture. The focus was given to paper and its reuse for functional parts of furniture. The goal was to manufacture upcycled board material, which could be used in furniture design in a similar way as other wood-based products, such as MDF or particleboards.

First, the project started with pressing different types of paper pulp (DIP and OCC) with starch and UF glue. These boards were manufactured and tested.

Finally, the used technology was high energy milling (HEM) of paper and pressing with hydraulic pressing machine with high temperature and pressure. When the boards were manufactured, the testing samples were cut according to EN 326-1 (1998). Its mechanical and physical properties were tested according to European Standards, namely: determination of density and density profile (EN 323: 2005); moisture content (EN 322: 2005); thickness swelling and water absorption (EN317: 2005); internal bond (EN 319: 2005); screw withdrawal resistance (EN 320: 2011); modulus of elasticity (MOE) and modulus of rupture (MOR) (EN 310: 1993).

4.1 Materials

On the following pages there are described the materials which were used during this research for the manufacturing the paper-based boards. The aim was to reuse paper and find a binder which can help to improve its physical and mechanical properties, so the boards can be optional material for regular industrially manufactured products. Several different materials and binders were tried and tested during this process.

The project of using paper started with using industrially processed paper pulp – deinked pulp (DIP) and old corrugated cardboard (OCC). The manufacture of the board started at Salzburg University of Applied Sciences, Campus Kuchl during the internship. The testing started without any additives, continued with starch adhesive and urea formaldehyde glue.

After first trials with samples of dimensions 20 x 20 cm and 20 mm thickness, the DIP and OCC was used for the testing. First tests were done without any additives – glue. The testing samples of wet paper pulp were pressed with single opening press Höffer HLOP 280 with pressure of 3,5 N/mm² and different temperatures – it started with 60 °C, 120 °C and finished with 180 °C.

After that the first samples with added glue were manufactured. They were bonded with two types of glue – starch adhesive and urea formaldehyde glue (UF). Both DIP and OCC were cut into testing samples and their physical and mechanical properties were tested. The second type of glue which was used was UF glue. The boards with thickness of 7 mm and 20 mm were manufacture.

Further development continued by using PLA as the binder for paper-based boards. Plastic was chosen because of similar way of manufacturing like in case of WPC boards. The testing samples of paper-PLA boards were done, and their physical and mechanical properties were also tested.

4.1.1 Paper used in the process

On the following pages there will be described the used types of paper for the experimental part of the research. The work has started with using recycled paper pulp, DIP and OCC. Later, the regular office paper and cardboard was collected from the waste centers and used for pressing the boards. At Figure 28 there can be seen the first trials with the paper pulp made by a kitchen blender and cardboard trays for apples, collected in the grocery store.

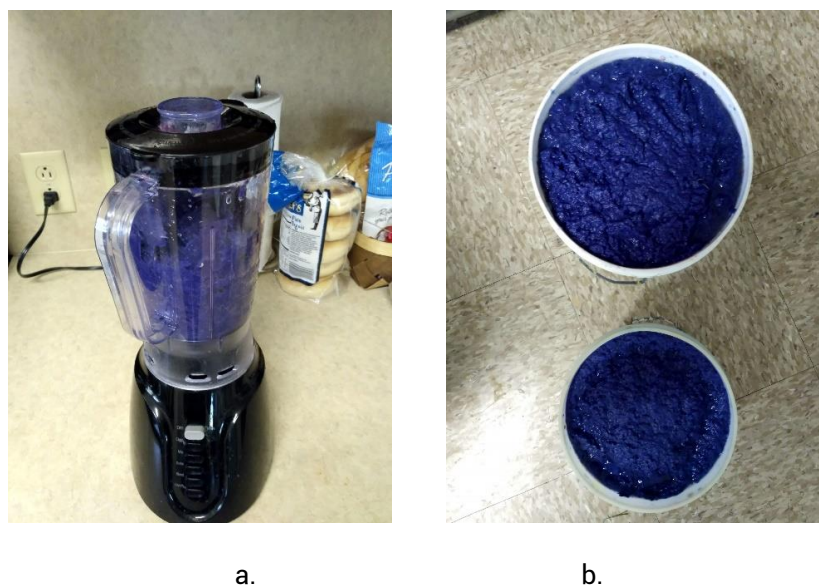


Figure 28 a. - Paper pulp first trials with a blender; b. – Blended paper pulp from cardboard trays for apples

4.1.1.1 Deinked pulp (DIP)

Deinked pulp (DIP) is recycled paper pulp made from old magazines, newspaper, and office paper. It is necessary to remove all chemicals, such as glue, pigments, and printing inks from the paper during the pulping process. The structure of the paper is changed due to this process and its fibers are weakened (Dukarska et al, 2018). The DIP which was used for the testing samples had the moisture content of 60 %. The used DIP can be seen at the following picture (Figure 29).



Figure 29 DIP used for the testing samples (author's archive, 2019)

4.1.1.2 Old corrugated cardboard (OCC)

OCC (Figure 30) is a paper pulp made from old, corrugated cardboard such as boxes and other packages. It is a post-consumer paper mainly used as a packaging material. It consists of a multiple layer structure. The recycling of OCC is straightforward if all non-paper materials are removed and its recycling also has several benefits. It reduces energy and water usage and the demand for untouched timber. OCC had also quite high moisture content which was about 60 %.



Figure 30 OCC paper pulp (author's archive, 2019)

4.1.1.3 Office paper (OP)

Office paper is a regular white paper used in most businesses for printing documents. Office paper is made of cellulose fibers. OP must be bleached and deinked to have its typical white color. This process lowers the physical properties of paper (Dukarska et al. 2018). For this reason, cardboard has better physical and mechanical properties which were confirmed during this research while testing the CB and OP paper-PLA boards. At the Figure 31 there can be seen the milled office paper prepared for the pressing in the wooden frame. The fraction of the paper was like the cotton wool. Its specific size was not measured. The office paper was milled two times together with the bioplastic PLA.



Figure 31 Office paper milled by HEM machine (author's archive, 2019)

4.1.1.4 Corrugated cardboard (CB)

Cardboard material was first created in 1870s as a material for the protection of shipped products. At first it was only a sheet of paper and in about twenty years the first double-faced paper was developed. It was and it is

still used mostly for folded boxes. Corrugated cardboard is a 100 % recycled material made from the recycled fibers of paper but also textiles without chemicals added because there is no need to remove printing inks or colors like the in case of office paper (Dukarska, et al., 2018). At Figure 32 there can be seen the milled cardboard paper by high energy milling machine. Cardboard was milled four times.



Figure 32 Cardboard paper milled by HEM machine (author's archive, 2019)

4.1.2 Adhesives

Adhesives have been used for ages. It started with the use of natural adhesives but during the 20th century they have been replaced with synthetic polymers. According to Wood Handbook (2010) a polymer is large molecule which consists of small units which are repeated. Synthetic polymers are either thermoplastic or thermosetting. Thermoplastics are long-chain polymers which can be reheated and they can harden again after cooling (Wood Handbook, 2010). Thermoplastics have generally smaller heat resistance and moisture resistance in comparison with thermosetting polymers. Thermosets, on the other hand, forms very strong structural adhesion, because of the chemical reaction which is irreversible. They form cross-linked polymers with high strength, resistance to moisture and chemicals. Thermosetting polymers are urea-formaldehyde, melamine-formaldehyde, phenol-formaldehyde, resorcinol-formaldehyde, iso-cyanate,

and epoxy adhesives (Wood Handbook, 2010).

Two types of adhesives were used during this research – starch adhesive and urea-formaldehyde (UF) adhesive. The main goal was to use natural based glue which would not affect the environment. UF glue is the most used glue in the board industry. The reason why it was used is, because of its comparison with industrially manufactured boards. In further development the PLA granulate was used as a binder of the paper fibers.

4.1.2.1 Starch adhesive

Starch is a natural, renewable, and biodegradable polymer produced by plants such as corn, rice, wheat, and potatoes (Gadhav et al., 2017). In the last years, starch became an attractive raw material, which could be used as wood adhesive thanks to its renewability and biodegradability, however the starch adhesives have not been applied more in an industrial scale due to its low mechanical properties and water resistance according to Yu et al. (2015). Starch adhesive is commonly used in the production of corrugated board and paper-based materials, packaging materials and wallpaper adhesive. The bond is developing while the water which must be added to activate the starch to stick, evaporates. Because the starch is a thermosetting polymer, it has a good heat resistance. Other substances can be added to improve its properties – for example UF glue for moisture resistance or PVAc to improve resistance to cold water. To inhibit the fungal growth, the biocides can be added (Emblem et al., 2012).

Starch is a white powder that swells due to hot water between the temperatures of 60 to 68 °C and it then becomes sticky. The biggest disadvantage of this glue is that they are not water resistant nor bacterial resistant.

4.1.2.2 Urea formaldehyde adhesive (UF)

Urea-formaldehyde adhesives are the most used in the wood-based products industry. Their typical use is for the manufacturing of particleboards, fiberboards, plywood, and other wood-based products. Ninety percent of particle boards is manufactured by using UF glue (Ormondroyd, 2015). They can form permanent, heat-resistant, they are low cost, resistant towards mold and they have excellent thermal properties. UF glue is a synthetic glue based on amino plastic resin. They belong to thermosetting plastic materials.

One of their disadvantages is that they can release formaldehyde which is carcinogenic. Due to these issues, there are the regulations for the use of UF glues, so they release less formaldehyde. They are supposed to be used in interiors only, because they lack the moisture resistance (polymerdatabase.com, 2020).

The used UF glue for this research was Leim plus 400 from Dynea while working on the thesis at SUAS. The glue is in the form of powder, which reacts after adding water. The temperatures of its use are between 70° 140 °C. Manufacturing of plywood, particleboard and fiberboard requires temperature of 210 °C. The properties can be seen at Figure 33 and in the Table 6.

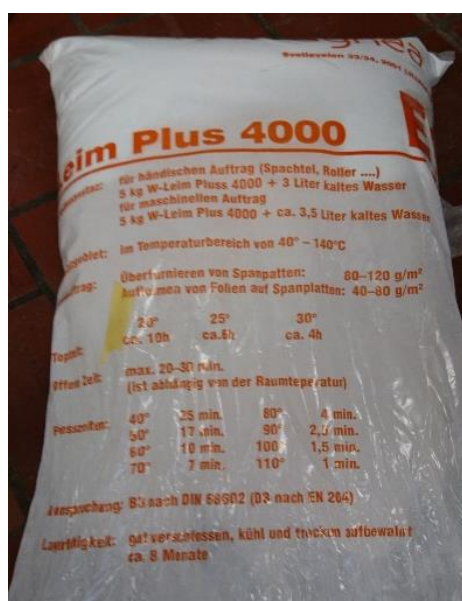


Figure 33 Used UF glue for pressing the boards (author's archive, 2019)

Table 6 General properties of UF glue (Dynea Leim Plus 4000)

Pressing temperature	40 ° - 140 °C
Pot-life	20 °C about 10 h 25 °C about 6 h 30 °C about 4 h
Gel time	20 – 30 minutes
Pressing times	40°C – 25 minutes 80°C – 4 minutes 50°C – 17 minutes 90°C – 2.5 minutes 60°C – 10 minutes 100°C – 1.5 minutes 70°C – 7 minutes 110°C – 1 minute

4.1.3 Binders and additives

During the research, the author also focused on the option of using plastic material as a binder. It was derived from the idea of manufacturing wood plastic composite boards (WPC), which are widely used as a substitute material for outdoor use. An example are terrace planks. Also, several additives are used in the production of wood-based products to improve its properties. One of the most common additives is wax, which provides the resistance towards water. Other additives which are used are fire retardants, moldicides or preservatives against fungi. On the following pages will be described the biodegradable plastic PLA and additive – limestone which were used in further development of paper-based boards.

4.1.3.1 *Poly lactide acid (PLA)*

To stay eco-friendly, the author used bioplastic granulate from PLA. According to Haider et. al, 2018, PLA is a biodegradable material, although it needs special composting conditions.

Biodegradable plastics became very popular materials ever since it was discovered that petroleum-based plastics have a negative effect on our environment. PLA was developed from natural materials to replace plastic based on petroleum for the environment protection. PLA is polyester produced from maize, wheat, sugar beets or agricultural waste. PLA is often used in biomedicine and food packaging (Panaitescu, et al., 2017). It is a thermoplastic and it can be processed like most thermoplastics into fibers, films or injection molded (Ashby, 2009). General properties of PLA can be seen in the following tables (Table 7, Table 8, Table 9, Table 10).

Table 7 General properties of PLA

Density	1210 – 1250 kg/m³
Price	2 – 4 USD/kg

Table 8 Mechanical properties of PLA

Young's modulus	3.45 – 3.83 GPa
Yield strength (elastic limit)	48 – 60 MPa
Tensile strength	48 – 60 MPa
Compressive strength	48 – 60 MPa
Elongation	5 – 7 %
Hardness –Vickers	14–18 HV

Table 9 Ecoproperties: material PLA

Embodied energy, primary production	52 – 54 MJ/kg
CO₂ footprint, primary production	1.23 – 1.55 kg/kg
Water usage	100 – 300 l/kg

Table 10 Recycling of PLA

Embodied energy, recycling	21.8 – 22.7 MJ/kg
CO 2 footprint, recycling	0.96 – 1.02 kg/kg
Recycle fraction in current supply	0.5 – 1 %

Typical uses: Packaging, filtering, writing; printing; currency; electrical and thermal insulation; gaskets.

Recycle mark

**Figure 34** Recycle mark for PLA (Ashby, 2009)

Typical uses. Food packaging, plastic bags, plant pots, diapers, bottles, cold drink cups, sheet, and film.

4.1.3.2 Limestone additive

To make the surface of the boards harder and more resistant, another additive which was used during the research is limestone powder. During the manufacturing process of wood-based products, limestone powder is often added into the boards such as particleboards or MDF boards to give them higher resistance towards water and strength. When using waste cardboard together with limestone powder to produce fiber cement boards, the flexural strength of cement composite boards can be improved by adding 10% of limestone powder (Khorami, et al., 2013). Limestone consists of petrified sea creatures. Its consistency is like chalk if it is porous or marble when it is fully dense. Most of the limestone consists also of sand (silicate) and other materials as well. Limestone is widely used as a building material and many old buildings such as the Georgia State Capitol was built with this material.

While choosing a purpose of the use of limestone it depends on its consistency and density. The soft one can be used for sculpting and decorative purposes (Granta, 2005).

4.2 Methods

The used methods during the research for the manufacturing of the paper based composite boards with eco-design rules were the milling of paper by HEM machine, pressing of paper boards by a hydraulic press and the testing of its physical and mechanical properties. The methods for the manufacturing of the boards are described in this chapter. The process of pressing the boards is like the manufacture of industrially manufactured wood-based products such as particleboards or MDF boards.

4.2.1 Manufacturing of paper-based boards

The used technology was pressing of paper together with different binders and the manufactured boards were tested in several different combinations of materials to create sustainable material, which could be used in furniture design. This part of research while using paper pulp was done at SUAS. It started with basic paper pulp which was received from Paper pulp recycling and milling facility in Austria. First attempts in manufacturing paper-based boards were made without any additives. Testing samples with dimensions of 200 x 200 mm were pressed with temperatures from 60°C, 120 °C to 180°C. After that additive such as starch adhesive and UF glue were added, and the boards were pressed over 180°C with the pressure of 3,5 MPa for 15 minutes. Its physical and mechanical properties were tested and compared with the industrially manufactured boards.

The research continued with the additive of PLA; biodegradable plastic based on corn starch described in the chapter 4.1.3.1. There were manufactured boards with thickness of 18 mm with temperature of 200 °C and pressing pressure of 3,5 MPa. The boards were pressed for 15 minutes. Mechanical and physical properties of these boards were also tested and

evaluated. In the following pages there will be described all the material combinations, its properties, and the best solution, which was used for furniture design. The research continued with the use of another plastic as a binder and testing of VOCs.

4.2.2 Pulping process

The traditional raw material for the pulping process is made of softwood such as spruce and pine or hardwood (beech wood, eucalyptus) and about 85 % is made of the materials mentioned above. About 10 % is then made of cotton linter (Sixta, 2006). While choosing the right pulping process for the raw material it is necessary to consider wood chemistry and composition to produce dissolving pulp (Chen et al., 2016).

There are two types of pulping processes, which are typically used:

- 1) *Acid sulfite (AS) process*
- 2) *Prehydrolysis kraft (PHK) process*

The AS process has been used for many years according to Sixta (2006). In this process the hemicelluloses, lignin and other components are removed from the wood chips and dissolved in spent sulfite liquor. Also bleaching of the pulp is another step (Chen, 2016). In comparison with AS, the PHK process consists of the combination of acidic (prehydrolysis) and alkaline (kraft) conditions. During the prehydrolysis stage, the hemicelluloses are extracted from the wood chips. Then comes the kraft cooking, bleaching of the pulp in order to get the desired purity of dissolved pulp (Sixta, 2006).

In this research there was obtained ready DIP (deinked pulp) and OCC (old corrugated cardboard) pulp from a pulping company in Austria.

4.2.3 High energy milling (HEM)

A crucial step in the manufacturing process of the paper boards is to mill the paper together with the plastic material. This process is done by High Energy Milling (HEM) which was done by the company FF Servis (Moravský

Beroun).

According to (Prokšán et al., 2017) the HEM is a type of milling with high energy which is transferred to the unit of the material. The process of HEM helps to mix the plastic throughout the paper fibers, so it has the right consistency. The manufactured boards by the pressing machine can be compared with the industrially manufactured MDF boards because of its similar structure, density, and properties.

4.2.4 Pressing of paper-based boards

The two different hydraulic presses were used since the research was done at SUAS and then it continued at MENDELÚ. Because of that this part is divided into two descriptions of the methods: Pressing of paper-pulp boards and Pressing of paper-PLA boards.

4.2.4.1 Pressing of paper-pulp boards

Pressing of paper-pulp boards was done while the pulp was still wet. The main technology which was used is pressing with single opening press (Höffer HLOP 280) at SUAS in case of paper pulp boards.

When pressing paper pulp boards, there were a different temperature during the process of pressing. The pressing started with temperature of 60°C, continued with 120 °C and 180 °C without any glue at first. After that the testing samples were made with different additives described in Materials part and the temperature of 180 °C. The pressing time was depending on the thickness of the boards. The first trials were 20 mm thick boards from paper pulp and starch adhesive. The pressing time in this case was 15 minutes. Next trial was a board with 7 mm thickness made from DIP and UF glue. The pressing time was 5 minutes. In case of 20 mm thick boards with UF glue it was 15 minutes.



Figure 35 Single opening press Höffer, SUAS (author's archive, 2019)

4.2.4.2 Pressing of paper-PLA boards

For the manufacturing of paper-PLA boards, the hydraulic press Stroztach (Figure 37) was used at the Department of Wood Science and Technology, MENDELU. The scheme of the hydraulic pressing machine can be seen at Figure 36. The paper-PLA boards were pressed with the temperature of 200 °C and pressure of 3.5 N/mm². To manufacture the paper-PLA boards, a high temperature of 200°C was used while pressing to melt the plastic so it can melt together. Four boards with the dimensions of 600 mm x 600 mm x 18 mm were manufactured from both CB and OP materials. The composition of each sample was composed of 2 kg of either CB or OP, 2 kg of PLA and 1 kg of limestone. The reasoning behind choosing these materials in combination with paper are described in the Chapter 4. Before the material could be placed in the hydraulic pressing machine, the material was pre-formed in a wooden frame with the dimensions of 600 x 600 x 18 mm. When the pre-forming step was done, the material was placed in between two bigger sheets of metal which

was then moved into the hydraulic pressing machine with the pressure of 3.5 N/mm² at a temperature of 200 °C for 15 min. The pressing time is always derived from the thickness of the board, which was 18 mm in case of paper-PLA boards. When the pressing was done, the boards were stacked vertically so they could cool down. They were conditioned by a relative humidity of 65 % and a temperature of 20 °C.

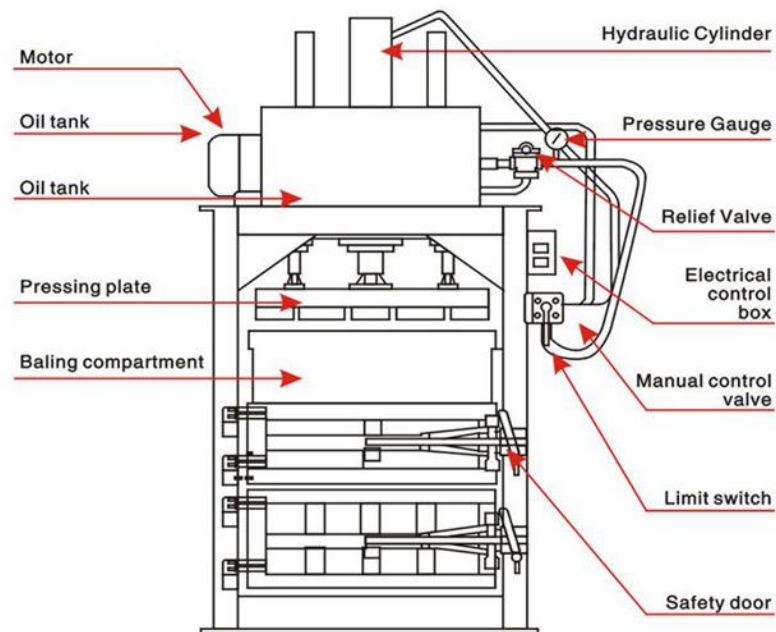


Figure 36 Scheme of hydraulic press (Machine MFG, 2019)



Figure 37 Hydraulic press Storztach, Department of Wood Science and Technology, Útěchov
(author's archive, 2019)

4.2.1 Testing methods of paper-based boards

Samples for the testing of physical and mechanical properties for both paper-pulp boards and paper-PLA boards were cut according to the European Standard EN 326-1 (1998). Most of the tests that were done for these boards because they are the most important for this type of boards, the dimensions of the testing samples are 50 x 50 mm. For the testing of MOR and MOE the dimensions are 50 x 410 mm which is counted this way:

$$l_2 = 20 \times \text{nominal thickness} + 50 \text{ mm} \quad (1)$$

l_2 – is the length of the testing sample

The testing of physical and mechanical properties was done according to the practices established for industrially manufactured boards and the usual properties that are tested. The most crucial properties for the boards used in

furniture design will be described. These are the properties which are crucial also for engineered wood products such as MDF or particleboards: Swelling in thickness and water absorption after 2h and 24 h of immersions in water (EN 317: 1993), Determination of density and density profile (EN 323: 1993), Modulus of Rupture (MOR) and Modulus of Elasticity (MOE) (EN 310: 1993), Internal Bonding (EN 319: 1993), Screw withdrawal resistance (EN 320: 2011).

4.2.1.1 Swelling in thickness and water absorption after 2h and 24 h of immersions in water (EN 317: 1993)

One of the crucial physical properties of industrially manufactured boards such as particleboards and MDF boards is swelling in thickness and water absorption. The reason is to know how resistant the material towards liquids is. This is very important to know at the board material used as a top desk for a table or surfaces like that. The testing of the specific samples (UF DIP 7mm) can be seen at Figure 38.

Thickness swelling (TS) [%] describes the increase of the sample thickness due to the water absorption. It is calculated as:

$$TS = \frac{t_2 - t_1}{t_1} \times 100 \text{ [%]} \quad (2)$$

TS is the thickness swelling [%]

t_1 Thickness of the test piece before immersion in water for 24 hours [mm]

t_2 Thickness of the test piece after immersion in water [mm]



Figure 38 Testing of swelling in thickness and water absorption of UF DIP samples (author's archive, 2019)

Water absorption WA [%] is calculated as:

$$WA = \frac{m_2 - m_1}{m_1} \times 100 \text{ [%]} \quad (3)$$

WA Water absorption related to weight [%]

m_1 Weight before the immersion inside the water for 24 hours [g]

m_2 Weight after the immersion in the water [g]

4.2.1.2 Determination of density and density profile (EN 323: 1993)

Another important property of industrially manufactured products like board materials is its density. Density of the boards helps the producer to predict how heavy the manufactured material for furniture or other uses will be. During the testing of both paper-pulp and paper-PLA boards their density was determined. Also, the density profile was measured in case of both tested materials.

The density ρ [kg/m³] is calculated using the following formula:

$$\rho = \frac{m}{w_1 \times w_2 \times t} \quad [\text{kg/m}^3] \quad (4)$$

m Weight [kg]

w_1 Width₁ [m]

w_2 Width₂ [m]

t Thickness [m]

4.2.1.3 Modulus of Rupture (MOR) and Modulus of Elasticity (MOE) (EN 310: 1993)

The panels or desk materials like particleboards or MDF boards and the researcher's paper-based boards should be resistant towards bending stress if they should be used on furniture. It is needed so the material will not be displaced when heavy objects are placed on the furniture parts. That is why it was necessary to test MOR and MOE of the boards according to the standards (EN 310, 1993 - Wood based panels. Determination of modulus of elasticity in bending and bending strength).

Modulus of elasticity in bending MOE [N/mm²]:

$$MOE = \frac{l_1^3 \times (F_2 - F_1)}{4 \times b \times t^3 \times (a_2 - a_1)} \quad (5)$$

MOE modulus of elasticity in bending [N/mm²]

l₁ distance between the centre of the supports [mm]

b width of the test piece [mm]

t thickness of the test piece [mm]

F₂ – F₁ increment of load in the straight-line portion of the load-deflection curve [N].

a₂ – a₁ the increment of deflection at the mid-length
of the test piece (corresponding to *F₂ – F₁*).

Modulus of Rupture MOR [N/mm²]:

$$MOR = \frac{3 \times F_{max} \times l_1}{2 \times b \times t^2} \quad (6)$$

MOR Modulus of Rupture [N/mm²]

F_{max} maximal load [N]

l, b, t length, width, thickness [mm]

4.2.1.4 Internal Bonding (EN 319: 1993)

The next very important property of wood-based products is the internal bond which is needed in order to have strong joints between each other especially if they are used for furniture from board materials like MDF boards or particleboards.

The tensions σ [N/mm²] were calculated according to the formula:

$$\sigma = \frac{F}{A_s} \text{ [N/mm}^2\text{]} \quad (7)$$

σ average tension across the surface [N/mm²]

F force applied from the testing device [N]

A_s sample surface [mm²]

4.2.1.5 Screw withdrawal resistance (EN 320: 2011)

A significant property of wood-based products for the use in furniture design is a screw withdrawal resistance (SWR). The purpose of testing this property is to find out if the material is strong enough to hold screws inside of them. The testing of SWR can be seen at Figure 39.



Figure 39 Testing of SWR of paper-PLA board (author's archive, 2019)

4.2.1 Testing methods of paper-pulp boards

During the first manufacturing and testing of the boards from DIP and OCC together with starch adhesive and UF glue 80 testing samples were made. For each property 20 samples were tested (MOR, MOE, IB, SWR, ST, D). The testing was done by the universal testing device Zwick/Roell Z 250. All the testing methods mentioned above were carried out, except of MOR and MOE in case of DIP and OCC starch adhesive boards. The reason was because of too high amount of moisture content inside of the material. During pressing these boards exploded. Because of that it was not possible to continue with the use of wet pulp while pressing. The dimensions of the testing samples were 50 x 50 mm for each sample (Figure 40). For the testing of MOR and MOE (EN 310:1993) the length of the testing samples was 20 times thickness of the testing samples plus 50 mm, which means 410 mm as described in 4.2.1. The samples were conditioned at a relative humidity of 65 % and a temperature of 20 °C.



Figure 40 Paper boards DIP and OCC with starch adhesive (180 °C, 120 °C, 60 °C) (author's archive, 2019)

4.2.2 Testing methods of paper-PLA boards

Further research continued with the use of OP and OCC materials described in the 4.1.1 and milled by HEM machine. The additive which was used as a binder was bioplastic (PLA). To give the boards higher resistance towards water and hardness, there was limestone added. The materials mentioned above were mixed with the additives and they were pressed with the thickness of 18 mm. The reason of this chosen thickness was because it is the most common thickness in furniture design. The final boards were supposed to be used for wood-based products such as a coffee table or storage furniture.

For each test 20 samples were made. In total there were 100 testing samples for each material (CB and OP). The samples were conditioned at a relative humidity of 65 % and a temperature of 20 °C. The testing of twenty samples for each property was done by the universal testing machine ZH050/TH 3A (Zwick Roell AG, Ulm, Germany). In the case of testing MOR and MOE according to EN 310 (1993) the dimensions of the testing samples were 50 x 410 x 18 mm. The testing samples for testing the internal bond according to EN 319 (1993) and screw withdrawal resistance EN 320 (2011) the dimensions were 50 x 50 mm. The testing of thickness swelling, and the water absorption was done in a laboratory with laboratory equipment which consists of the beakers, digital caliper, and laboratory scale.

4.2.3 Testing the emission of VOCs – paper-PLA boards

Testing of the emissions of VOCs was done only in case of paper-PLA boards, since they were used for the final design of furniture presented in this thesis (Figure 41).

The measuring was done with sampling pump Gilian LFS – 113 and assessed using Field and laboratory emission cell (FLEC). Air volume flow was 12 l/h where the time of sampling the air is 180 minutes. The measuring was done according to (ČSN EN ISO 16000-9, 2007).

Emission collection from building materials can be done in the Field and laboratory emission cell (FLEC), when the testing sample is stored in the testing cell (according to the size of the sample). After that, the offtake is done by the sorption tube. Because of the need of method for easy identification of VOCs in building materials in place, the terrain and laboratory emission chamber (FLEC) were done, and it became European standard for the testing of emissions (Zhang, 2003).

If the measuring sample is material with smooth surface, the emission chamber is placed immediately onto the surface of the testing sample like in case of paper-PLA boards. Other materials must be placed into special equipment of the testing sample, which provides tightness.

The testing was done in case of the boards with following dimensions:

- Ø 450 mm, thickness 18 mm (CB; 1 piece)
- Ø 355 mm, thickness 18 mm (OP; 1 piece)
- 500 x 500 mm, thickness 8 mm (CB; 2 pieces)

The testing is done with the current climate conditions where the temperature was 23 °C and relative humidity was 50 %. The air is brought to the chamber, and it cannot consist high values of VOC. Concentration of any substances have to be lower than 2 µg.m⁻³ and the total TVOC (total volatile organic compound) cannot exceed 20 µg.m⁻³. That is why the so cold blank test is done before the measuring of the sample, when the emission chamber is placed on the inert glass (ČSN EN ISO 16000-10, 2007).

After the testing emission chamber is turned on, it is stabilized after 20 minutes to stabilize the whole system. The carrier gas (nitrogen) is blown through the whole system. In the 1st step the blank test is measured from inert glass on which the sample is placed to check the cleanliness of the chamber. In the 2nd step the testing sample is placed in the testing chamber. Two desorption tubes are attached to the chamber and the pump is screwed on them as well. The carrier gas is entering into the system at the flow speed of 250 ml.min⁻¹. Behind the chamber there are placed two pumps with the flow

speed of 100 ml.min^{-1} and the rest flow speed of 50 ml.min^{-1} is being evaporated into the room. The offtake is being done during 3 hours with the amount of 18 liters on each pump. After the offtake is done, the data is analyzed with the chromatograph.



Figure 41 Testing the VOCs of OP (author's archive, 2020)

5. Results and discussion

In this chapter of the thesis, the results from the whole research will be described. During the research, the goal was to map the current situation in furniture design while taking sustainability in mind and increase the awareness about designing in a sustainable way.

Thanks to the previous research the author has come to the idea of reusing wastepaper in her own manufacturing process of paper-based boards, which can be used for furniture design or other wood-based products. The results of the manufacturing and testing boards made of salvaged material will be described in this part – waste post-consumer paper (DIP, OCC, OP and CB) which was one of the goals of this thesis. The physical and mechanical properties of both paper-pulp boards and paper-PLA boards will be described and explained.

One of the goals was to find the possible combination of the materials which could be used for the manufacturing of the boards used in furniture design project, designed by the author of the thesis. As seen in the Literature review, there are many new approaches in the material engineering and designers are looking for new ways of creating new products. In a similar way as the mentioned company Dear Human (3.11.7.3) from Canada, which is working with wastepaper, the author of the thesis was using as a main source the recycled paper. The idea was to reuse wastepaper collected from waste centers, because not all the paper can be recycled. Many times, the sorted waste materials end up in a landfill or they are incinerated. As Shalbafan et al. (2016) did in their research where they used wood and plastic, the authors idea was a bit similar, although while using paper instead of wood.

On the following pages, the testing of physical and mechanical properties of the manufactured paper-based boards will be described as the result of the research on the topic of Sustainability in Furniture Design. Also, the results of the testing of the emission of VOCs in the paper-PLA boards will be described in this part.

All the results were taken into consideration while using the paper boards for wood-based products designed as part of this thesis. The most suited option of material combination was used for the desktop of coffee tables. The results were presented at the international conference in *“Wood Science and Engineering in the third millennium”* (ICWSE) 2019 in Brasov, Romania (see Appx.2). An article was published in the proceedings from the conference with the topic *“Furniture design with principles of sustainable design by using paper pulp”* (Sobotková, et al., 2019) and then also published in a special number of scientific magazines called ProLigno 2019. Vol. 15. nr. 4 in December 2019. Other results with the name *“Paper and Plastic Waste Composite Boards for Furniture”* were presented as an abstract at the *“Society of Wood Science and Technology International Convention”* in Portorož, Slovenia in 2020 (held online due to the Covid-19 pandemic) (see Appx.3). Another part of the research was published in the scientific impact journal - Bio Resources with the topic of *“Mechanical and Physical Properties of Boards Made from Recycled Paper”* (Sobotková, et al., 2021) (see Appx.1).

5.1 Results for paper-pulp boards

The results of testing the mechanical and physical properties will be described in this part. The results of each testing method which were described in chapter 4.2.1 were compared with industrially manufactured MDF boards. Although this material should not substitute the industrially manufactured boards, but it is designed as another option of what could be used in wood-based products but possibly even other areas in the constructions of interior design.

5.1.1 Physical properties

Physical properties of the pulp boards are described in this part. The physical properties which were tested are Density profile according to EN 323 (2005) and thickness swelling and water absorption according to EN 317 (2017).

5.1.1.1 *Density profile of paper-pulp boards*

One of the evaluated properties was the density of the boards. In the graphs below (Figure 42, Figure 43, Figure 44, Figure 45) can be seen the density profile of the samples made with different temperatures, starting at 120 °C, 180 °C without any glue and continued with the starch adhesive and temperature of 180 °C. There has been significant difference between the boards made without any glue and with the glue added. All the results will be described at the following pages.

In the graph below (Figure 42) can be seen that the testing samples without any additives and pressed with the temperature of 120 °C and 180 °C had the lowest density which was around 450 kg/m³. These results were expected, although the reason why the boards were tested was to find out how the material behaves during the pressing. There are also the results of the boards with starch adhesive pressed with 180 °C. All these samples had 20 mm thickness. Samples with added starch adhesive had significantly higher density where the average value was around 700 kg/m³. The highest density had the 180 UF DIP 7 mm boards, which was 1078,67 kg/m³.

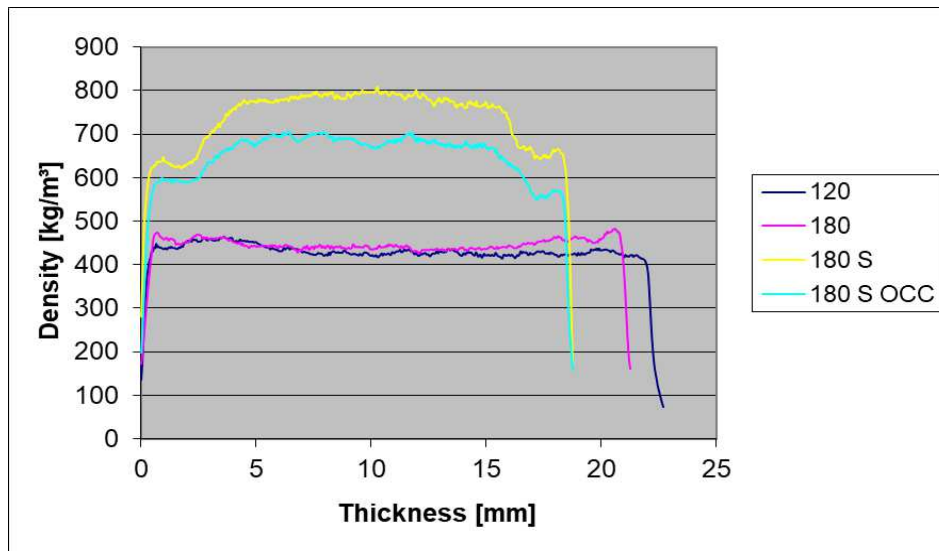


Figure 42 Density profile of boards with starch adhesive 20 mm thickness

In case of the samples with UF glue, 20 mm thickness and the pressing temperature of 180°C had slightly higher density than the starch adhesive boards. The following graph (Figure 43) is showing that the DIP and OCC samples with UF glue had the average density of 800 kg/m³. The reason why those samples had better results than the starch adhesive samples (S DIP, S OCC) is because of the mechanical properties of these types of glue, which is described in the Chapter 4.1.2.

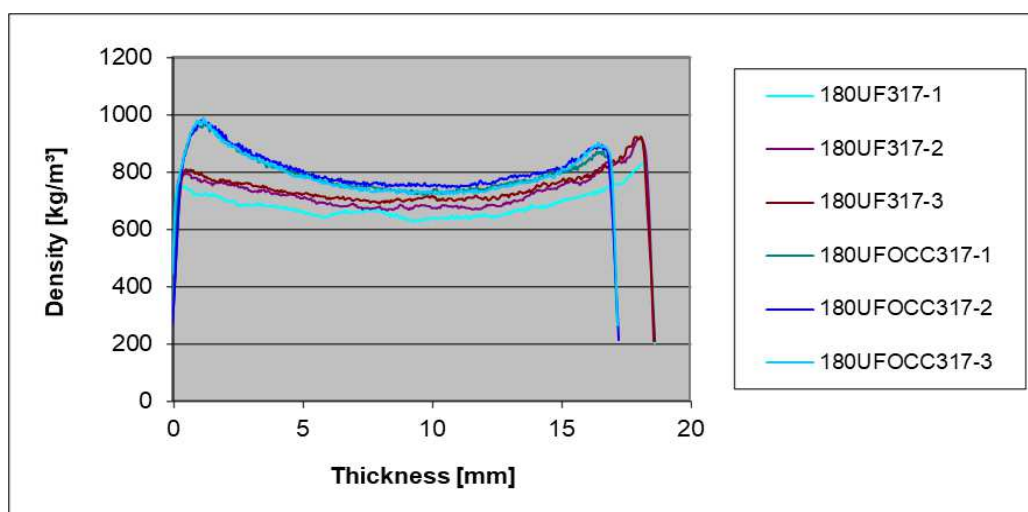


Figure 43 Density profile of boards with UF glue 20 mm thickness

The highest density of the board was in case of 7 mm thick samples with UF glue. The average value was 1200 kg/m³ which can be seen at Figure 44. These boards had better results because of its thickness – we can see that the material was pressed more which caused higher density.

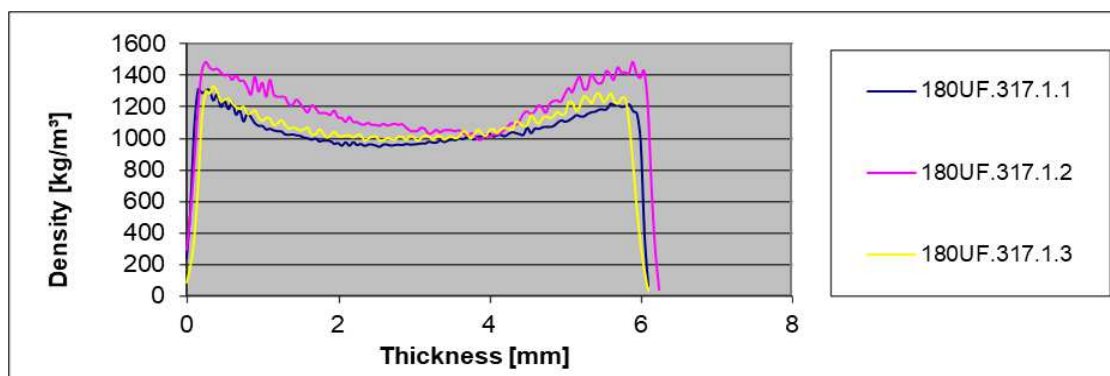


Figure 44 Density profile of boards with UF glue 7 mm thickness

5.1.1.2 Thickness swelling and water absorption after immersion in water for 24h of paper-pulp boards

Another important property for such boards is swelling in thickness and watery absorption. The testing samples with the dimensions of 50 x 50 mm of each sample, which were previously cut according to the standard (EN 317: 1993), were immersed into the water for 24 hours (Figure 45). The dimensions and weight of each sample were measured before the immersion into the water and then the same was done after 24 hours of immersion. With these values the swelling in thickness and water absorption could be calculated according to the following formulas: (2), (3) mentioned in 4.2.1.

The result of each sample is described in the Table 11 where are also presented the results of the density profile. According to the calculated results, the lowest thickness swelling had the OCC sample with starch adhesive, which was 19,97 %. On the other hand, the worst results had the 7 mm thick UF sample. We can observe that the thinner the board the water can enter inside more which is causing higher swelling in thickness.



Figure 45 Swelling in thickness (author's archive, 2019)

Table 11 Physical properties of paper-pulp boards

Testing method	Density profile EN 323	Swelling in thickness EN 317
material/thickness/units	[kg/m ³]	[%]
120 DIP 20 mm	422,30	31,12
180 DIP 20 mm	439,76	27,90
180 S DIP 20 mm	730,59	22,52
180 UF DIP 20 mm	841,96	29,50
180 S OCC 20 mm	641,16	19,97
180 UF OCC 20 mm	659,10	21,03
EGGER MDF ST E1 >19 - 30 mm	600 - 800	<10
180 S DIP 7 mm	-	-
180 UF DIP 7 mm	1078,67	52,93
EGGER MDF ST E1 >6 - 9 mm	600 - 800	<17

All the paper-pulp boards were compared with industrially manufactured MDF boards (Egger). We can say that the density is very similar to the MDF boards. On the other hand, the swelling in thickness is worse than in case of MDF boards – the value should be less than 10 % in case of industrially manufactured boards. In case of paper-pulp boards the lowest value had S OCC 20 mm thick testing sample with the value of 19.97 %. It is caused by no surface finish of the paper-pulp boards. That is why the water can enter the surface very easily and cause the swelling.

5.1.2 Mechanical properties

Mechanical properties are important for the manufacturing process of wood-based products. The mechanical properties which were tested are the internal bond (IB) according to EN 319 (2005), modulus of rupture (MOR) and modulus of elasticity (MOE) were tested according to EN 310 (1993) and screw withdrawal resistance (SWR) according to EN 320 (2011). These properties are crucial for the boards used for furniture design and wood-based products. They were calculated according to the formulas mentioned in the chapter 4.2.1 – (5), (6), (7).

In the table below (Table 12) there are all the results of testing the mechanical properties of samples compared with industrially manufactured boards from EGGER with the corresponding thickness. Unfortunately, it was not possible to measure MOR and MOE of the boards with the 20 mm thickness while using starch adhesive, because it was not possible to manufacture the boards with the right dimensions due to the high moisture content of the boards. These boards exploded while pressing.

5.1.2.1 MOR and MOE of paper-pulp boards

On the following pages there are presented the results of testing the MOR and MOE which was possible to test only in case of the UF DIP 20 mm boards, UF OCC 20 mm boards and UF DIP 7 mm boards.

The UF glue 20 mm boards had following average values for DIP - MOR of the boards was 6.13 N/mm² and MOE was 1189.07 N/mm². These tests showed that the boards are not very resistant towards bending and does not have big bending strength. In comparison with the industrially manufactured boards where the value of MOR should be higher than 18 N/mm² the MOR was quite low which stands also for the MOE – that should be over 2100 N/mm². The reason could be high moisture content of the boards before pressing. In case of UF OCC 20 mm boards the MOR was 5.89 N/mm² and MOE was 750.83 N/mm². Those results were even worse than in case of UF DIP 20 mm boards. OCC material could be less mixed than the DIP material.

The UF DIP 7 mm thick board had the MOR of the boards 14.29 N/mm². The MOE of these boards was 3019.93 N/mm². Compared to the industrially manufactured boards the MOE is higher than the standard value which should be more than 2700 N/mm². On the other hand, the MOR did not exceed the standard value which should be more than 23 N/mm². Overall, the board with the 7 mm thickness had the best results from those testing samples.

MOR and MOE of the paper-pulp boards with the starch adhesive could not be tested due to the problems with pressing the boards with the needed length for these tests due to the amount of moisture content. That is why none of the following properties could not be tested and they are not presented in the thesis.

5.1.2.2 Screw withdrawal resistance of paper-pulp boards

The testing of SWR proved how important is the chosen additive of the boards. We can observe that the boards without any glue had much lower SWR than the boards with either starch or UF glue. The highest SWR had the UF DIP 20 mm boards with the value of 1899.97 N. Compared to the industrially manufactured boards it is a higher value. SWR should be higher than 1080 N. Another testing samples which had higher SWR than the average of industrially manufactured boards were the samples of UF OCC 20 mm board which had a value of 1277.94 N. According to these results we can say that the UF glue is providing much better resistance towards the screws and the testing samples are much stronger.

In case of the starch adhesive boards the average value was lower than the standard value. The samples S DIP 20 mm had the average value of 1031.93 N. Screw withdrawal resistance of S OCC 20 mm boards had the average value of 953.35 N. We can observe that the starch adhesive has lower mechanical properties than the UF glue as described in the 4.1.2.

We can say that the samples with the higher thickness of 20 mm have better results of SWR than the samples with the 7 mm thickness. The UF DIP boards had the SWR average value of 542.95 N. On the other hand, the boards which were manufactured without any glue and with a lower temperature had very low values could not be used for furniture design and stressed parts, but they might have a good use for a wall panel, where the strength is not so important.

Overall, the best tested results were those of the 20 mm thick boards made of DIP and UF glue. In all the results we can observe the descent of the resistance with either starch adhesive or without any glue and with the smaller thickness.

5.1.2.3 Internal Bond of paper-pulp boards

The internal bond (IB) was another important property to test. The results were also compared with the industrially manufactured boards, where the value should be higher than 0.55 N/mm^2 . The tested results showed that the boards do not exceed the recommended value, which was probably caused by the high moisture content mentioned in the chapter 4.1.1.

The closest results had the UF OCC 20 mm boards with the value of 0.51 N/mm^2 . Comparable result had also the 7 mm thick UF DIP board with the value of 0.41 N/mm^2 . Following result had the UF DIP 20 mm board with the value of 0.39 N/mm^2 . The boards from starch adhesive and DIP with 20 mm thickness had the value of the internal bond 0.37 N/mm^2 . The boards without any additives had insignificant value. According to those results we can say that the added glues are giving the boards significantly higher strength.



a.



b.

Figure 46 a- Testing of Internal Bond; b – Testing of MOR (author's archive, 2019)

Table 12 Comparison of testing the mechanical properties for DIP and OCC boards (7 mm and 20 mm thickness)

TESTING METHOD	IB	SWR	MOR	MOE
	EN 319	EN 320	EN 310	EN 310
material/thickness/units	[N/mm ²]	[N]	[N/mm ²]	[N/mm ²]
120 DIP 20 mm	0,01	182,26	-	-
180 DIP 20 mm	0,01	127,08	-	-
180 S DIP 20 mm	0,37	1031,93	-	-
180 UF DIP 20 mm	0,39	1899,97	6,13	1189,07
180 S OCC 20 mm	0,33	953,35	-	-
180 UF OCC 20 mm	0,51	1277,94	5,89	750,83
EGGER MDF ST E1 >19 - 30 mm	>0,55	>1080	>18	>2100
180 S DIP 7 mm	-	-	-	-
180 UF DIP 7 mm	0,41	542,95	14,29	3019,93
EGGER MDF ST E1 >6 - 9 mm	>0,65	-	>23	>2700

5.1.3 Discussion – paper-pulp boards

The testing of paper-pulp boards was a great start for the development of such paper-based board material which could be used in furniture and interior design or simply for wood-based products. The first tests gave an idea of the behavior of paper pulp with the moisture content about 60 %. There were manufactured boards from DIP and OCC with starch adhesive and UF glue to compare its properties, even though the main goal was to use natural or biodegradable glues. The UF glue is the most common type of glue in the engineered manufactured products as such boards (MDF, particleboards) so it was used to compare the results of both additives used.

During the process of testing there have been some errors with the manufacturing which also brought very valuable knowledge to the research. Due to the big amount of moisture inside the paper pulp, which is a natural substance during the pulping process it was not possible to press all the testing samples. The bigger the dimensions of the intended samples, which means the surface, the bigger the problem with the steam, which was coming out of the boards. Since the samples for testing the MOR and MOE had to have the length of about 25 cm it was impossible to manufacture them, because the steam coming out caused an explosion while pressing. The bending strength was tested only for UF DIP and OCC 7- and 20-mm boards where the best results had UF 7 mm boards. In comparison with DIP and OCC, the DIP 20 mm boards had better results. MOR was 6.13 N/mm^2 and MOE was 1189.07 N/mm^2 . The best results were the 7 mm boards where the UF glue was used and had the best results during the testing. These boards were compared with industrially manufactured boards from EGGER MDF ST E1 >6 - 9 mm. The 7 mm UF boards had good MOR and MOE, which can be caused by its high density. These boards had the highest screw withdrawal resistance which was 1899,97 N. It also had the highest density in comparison with other boards - $841,96 \text{ kg/m}^3$.

Unfortunately, the starch adhesive boards with the thickness of 7 mm were not possible to manufacture, because of the high moisture content of 60 % inside of the material. These boards also exploded while pressing. The pressing of such moisturized boards was possible only in the case of smaller testing samples with the dimensions about 250 x 250 mm.

Even though there were some errors while manufacturing the first samples, there were also very good results during the process which helped to the further development. In general, the swelling in thickness was higher than in comparison with the industrially manufactured MDF boards from EGGER (EGGER MDF ST E1 >19 - 30 mm) boards, where the value should be lower than 10 %. The best swelling in thickness had OCC samples with the starch adhesive where the value of swelling in thickness was about 19.97 %.

The best density had the boards of the 7 mm thickness made of DIP and UF glue where the density was 1078.67 kg/m³ where in comparison with the industrially manufactured boards from EGGER MDF ST E1 >6 - 9 mm should be between 600 – 800 kg/m³.

The internal bond of 0,51 N/mm² had OCC 20 mm UF board which is very good result in comparison with the industrially manufactured boards by EGGER MDF ST E1 >19 - 30 mm which should be higher than 0,55 N/mm². After the first manufacturing and testing the samples was done the author was planning to do further testing with another additive, where it would not be possible to use any glue.

5.2 Results for paper-PLA boards

As described in the previous parts, further development of the research continued with another additive which was used instead of glue. Further development of the paper-based boards brought an idea of using a bioplastic in a similar way as in case of WPC boards as described in the 3.11.3. The used material was a biodegradable bioplastic PLA together with the additive of limestone powder as it is used during the manufacturing process of particleboards as described previously (4.1.3.2).

The amount of the materials was in the ratio of 2:2:1 of paper, PLA, and limestone – 2 kg of paper (CB or OP), 2 kg of bioplastic (PLA) and 1 kg of limestone powder. Both materials – CB and OP were milled by HEM machine together with PLA and limestone powder. The advantage of this process is that even smaller fibers of paper could be reused. In the process of recycling paper, it is a great advantage because even recycled products such as cartons for eggs or trays for fruit which are usually made from recycled paper pulp could be reused. Theoretically, at the end of the lifetime of these boards, they can possibly be broken down and milled again. This would be an example of circular manufacturing and it follows the rules of Circular Economy described in chapter 3.2 – *Circular economy*, although the paper fibers would get smaller each time of milling which would lower its strength. At this point comes another opportunity for testing at this research.

After the milling of all material, two boards from each material with dimensions of 600 x 600 mm were manufactured and its physical and mechanical properties were tested according to European standards as well.

Overall, CB had better results in comparison with OP. It was probably caused by the amount of milling of the material which was four times more than in case of CB and only twice in the case of OP.

For each test there were 20 testing samples which means there were 40 testing samples for physical properties and 60 testing samples for mechanical properties. In total there was 100 testing samples.



Figure 47 Paper-plastic boards (CB and OP) (author's archive, 2019)

5.2.1 Physical properties

In this part the physical properties of the paper-PLA boards will be described. In total 20 testing samples were needed for the testing – 20 samples for swelling in thickness and water absorption and 20 samples for testing the density. The results for the density, thickness swelling, and water absorption tests can be seen in the tables below (Table 13 and

Table 14). Testing of physical properties was done according to European standards, namely thickness swelling and water absorption after 2 hours and 24 hours of immersion in the water EN 317 (1993).

5.2.1.1 Density profile of paper-PLA boards

The density profile can be seen at the Figure 48 and Figure 49. The density of the CB boards was between 632.2 and 837.8 kg/m³, while the density of OP ranged from 655.7 to 873.1 kg/m³. The density of the paper-PLA boards can be compared with industrially manufactured particleboards, because of its similar manufacturing process and its properties. The authors boards created during this research can be compared with the two types of particleboards used in different conditions according to EN 312 (2010) *"Particleboards: Specifications"*. Basically, the particleboards are divided in the types of boards such as Type P2, P3, P5, P6. The paper-plastic boards can be compared with the Type P2 - Boards for interior fitments (including furniture) and Type P3 - Non-load bearing boards for use in humid conditions. The CB and OP materials had average values of 749.0 and 779.3 kg/m³. The OP board had higher variance of the values in case of the density which was caused by the variations in the milling times of the material. The milling times was different because of different consistency of each material – CB has naturally bigger particles than OP, which has already lowered physical and mechanical properties by the bleaching and deinking process as described in the 4.1.1. That is why OP was milled twice and CB four times. However, this caused that the bioplastic PLA was not spread equally between the paper as in case of CB (Sobotková, et al., 2021).

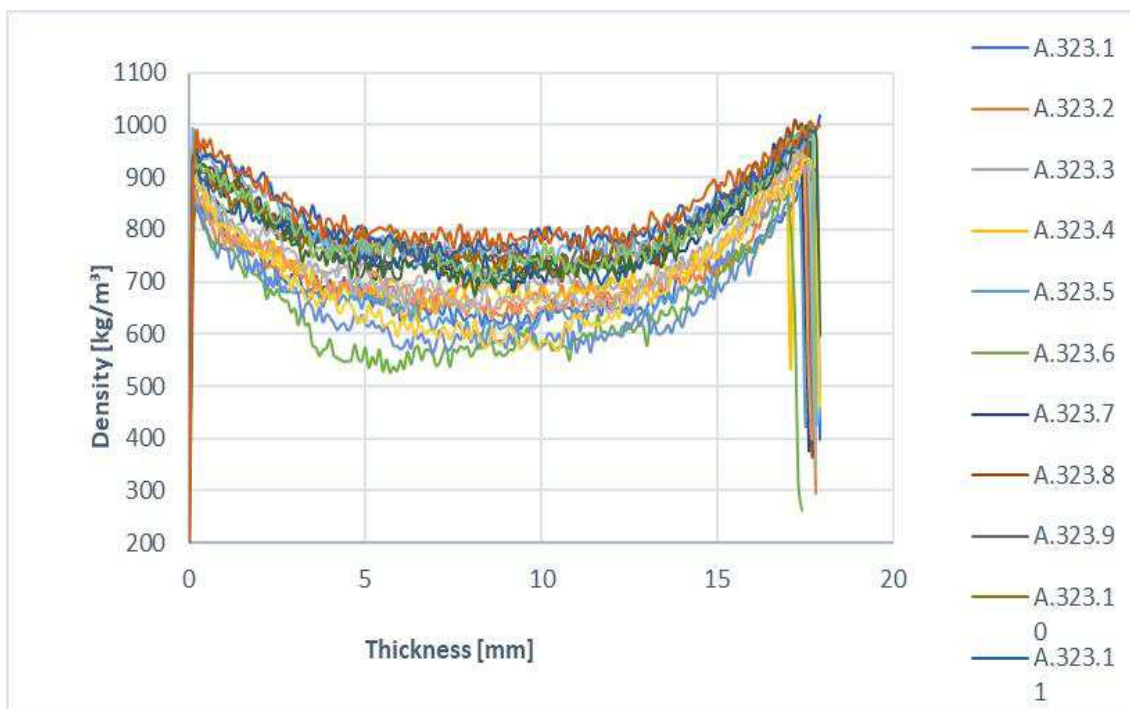


Figure 48 Density profile of CB board, 18 mm

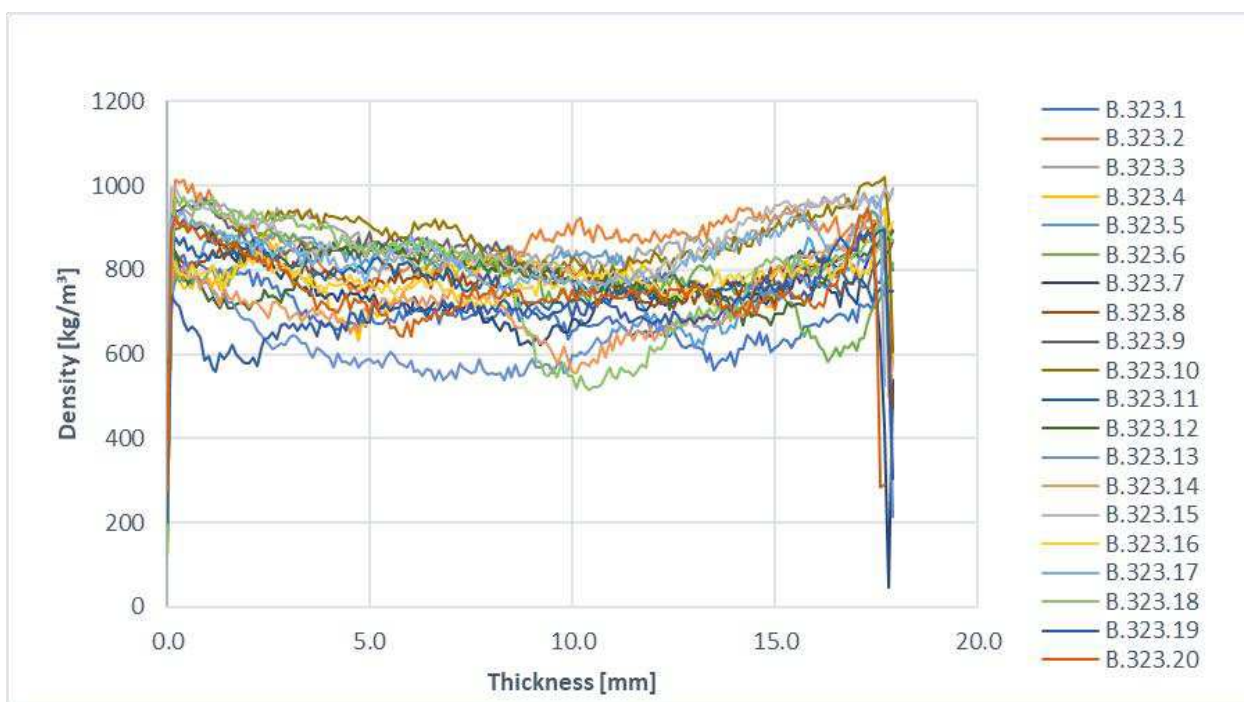


Figure 49 Density profile of OP board, 18 mm thickness

5.2.1.2 Thickness swelling after immersion in the water for 2 hours and 24 hours of paper-PLA boards

Swelling in thickness was also calculated for both samples – CB and OP after 2 hours at first and 24 hours after that. The thickness swelling of the CB was between 0.43% and 4.44% after 2 h of immersion in the water and 1.78% to 5.96% after 24 h of immersion in the water. The thickness swelling of the OP was between 3.54% and 9.73% after 2 h of immersion in the water and 5.11% to 10.61% after 24 h of immersion in the water. The average value for CB was 3.73% and 7.64% for OP samples. The results were then compared with the limits for the Type P3 boards where the limit is 13 % according to EN standard 312 (2010). The water absorption resistance was caused by the added plastic components together with the high-speed milling cycles. Even though OP had lower values, both board materials had still very good swelling in thickness results compared to the general requirements (Sobotková, et al., 2021).

The results were also analyzed statistically. A one-way factor ANOVA was used to analyze the relationship between the swelling in thickness and the tested samples of CB and OP. The CB had a significantly lower swelling in thickness value than the OP material, as can be seen in Figure 50. The reason is probably because there was a higher density of the CB material and the bigger strength of CB fibers, which has led to much stronger boards (Sobotková, et al., 2021).

There was a higher difference between the CB and OP in the case of the water absorption. It was 15.97 % in case of CB and 64.94 % in case of OP. The reason why there is this difference is again because of the natural properties of CB, which were described in the chapter 4.1.1.2. In general, CB material is much stronger and more resistant than regular OP material, weakened by the bleaching and deinking process.

Table 13 Density, swelling in thickness and water absorption of CB statistics

Testing Method/CB	Mean	Geometric Mean	Minimum	Maximum	R	Median	Variance	Standard Deviation
Density (kg/m ³)	749.00	746.86	632.25	837.75	205.49	749.00	3134.98	55.99
Thickness Swelling 2 h (%)	1.74	1.47	0.43	4.44	4.01	1.67	0.99	0.99
Thickness Swelling 24 h (%)	3.73	3.61	1.78	5.96	4.18	3.75	0.81	0.90
Water Absorption 2 h (%)	6.05	5.79	4.18	12.16	7.98	5.34	4.06	2.02
Water Absorption 24 h (%)	15.97	15.67	11.13	22.71	11.58	15.63	9.90	3.15

Table 14 Density, swelling in thickness and water absorption of OP statistics

Testing Method/OP	Mean	Geometric Mean	Minimum	Maximum	R	Median	Variance	Standard Deviation
Density (kg/m ³)	779.34	781.89	655.74	874.69	218.95	776.73	3214.94	56.70
Thickness Swelling 2 h (%)	6.10	5.95	3.54	9.73	6.18	5.50	2.82	1.68
Thickness Swelling 24 h (%)	7.64	7.47	5.11	10.61	5.50	7.70	2.14	1.46
Water Absorption 2 h (%)	49.07	47.44	27.54	73.72	46.18	49.61	171.59	13.10
Water Absorption 24 h (%)	64.94	63.30	47.35	82.41	35.06	64.48	101.30	10.06

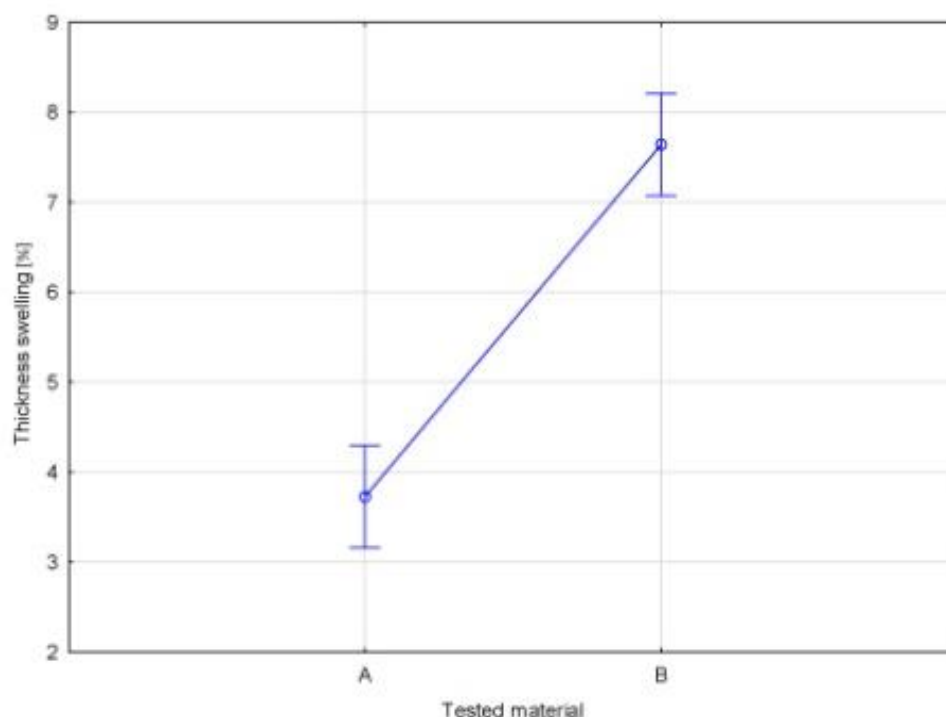


Figure 50 One-way factor ANOVA for Thickness swelling and tested materials

5.2.2 Mechanical properties

For the use in furniture design and such products and especially for the use on the desk-tops or tabletops, the crucial mechanical properties are flexural. Mechanical properties were tested according to European standards, namely internal bond EN 319 (1993), MOR and MOE EN 310 (1993), screw withdrawal resistance EN 320 (2011). All the results were compared with the standard EN 312 (2010), where each group of the types of boards is described. For each test 20 testing samples were used. In total there were 60 testing samples for mechanical properties as mentioned before.

5.2.2.1 MOR and MOE of paper-PLA boards

As mentioned before, the boards can be compared with the Type P2 with the requirements for boards for interior fitments (including furniture) where the value of MOR for boards with thickness between 13 to 20 mm is 11 N/mm² and MOE has the value of 1600 N/mm². The MOR of CB ranged between 5.02 to 10.82 N/mm², with an average value of 8.45 N/mm². The

MOR of the OP ranged from 1.98 to 6.57 N/mm², with an average value of 4.91 N/mm². The MOE of CB ranged from 1215.0 to 2231.5 N/mm², with an average value of 1931.3 N/mm².

The OP boards can be compared with the requirements for general purpose boards for the use in dry conditions (Type P1), where the MOR is 10 N/mm². The MOE of the OP ranged from 582.9 N/mm² to 1992.5 N/mm², with an average value of 1529.2 N/mm² (Sobotková, et al., 2021). CB had the value of MOE 1931.3 N/mm², while OP had the value of MOE 1529.2 N/mm². The reasoning behind this difference is caused by the nature of CB, which is in general much stronger than OP materials as described in the chapter 4.1.1.

5.2.2.2 Screw withdrawal resistance of paper-PLA boards

Following testing was the testing of SWR. The testing was done on both the face of the boards and the edge of the boards. The testing on the face had the value of 316.6 N for the CB material and 347.2 N for the OP material, while the edge resistance was 170.9 N for the CB material and 160.6 N for the OP material. The boards were compared with industrially manufactured boards such as MDF boards where the resistance for an 18 mm board is higher than 1080 N (EN 320: 2011). The results for the paper-PLA boards were much lower. The reason can be attributed to the mechanical properties of paper in general, which was described in the material part of the thesis as mentioned earlier in this work.

5.2.2.3 Internal bond of paper-PLA boards

Other crucial properties for such boards is the internal bond. The internal bond test values of the CB and OP samples ranged from 7.70 to 3.53 N/mm², however, the CB samples had a larger range than the OP samples. The internal bond of CB samples ranged from 1.90 to 13.35 N/mm² and 1.33 to 6.55 N/mm² for the OP samples (Sobotková, et al., 2021). The minimum internal bond value for boards with a thickness of 18 mm according to the EN standard 312 (2010) that are used for interior fitments (including furniture) in dry conditions (Type

P2) is 0.35 N/mm². The reasoning behind this is again the natural properties of cardboard which has bigger fibers and glue was already added, while office paper is bleached to have white color and the fibers thereby lose some of their strength.

Table 15 Mechanical Properties of the CB

CB								
Testing Method	Mean	Geomet. Mean	Minimum	Maximum	R	Median	Variance	Std. Dev.
MOR (N/mm ²)	8.45	8.24	5.02	10.82	5.80	8.62	3.09	1.76
MOE (N/mm ²)	1931.32	1902.75	1215.05	2231.47	1016.42	2047.49	94365.23	307.19
Internal Bond (N/mm ²)	7.70	6.90	1.90	13.53	11.63	7.10	10.30	3.21
SWT- Surface (N)	316.59	298.38	150.12	581.44	431.32	284.43	12277.43	110.80
SWT- Edge (N)	170.93	149.35	23.77	366.04	342.27	169.43	5813.25	76.24

Table 16 Mechanical Properties of the OP

OP								
Testing Method	Mean	Geo. Mean	Min.	Max.	R	Median	Variance	Std. Dev.
MOR (N/mm ²)	4.91	4.71	1.98	6.57	4.59	5.32	1.38	1.17
MOE (N/mm ²)	1529.22	1458.89	582.90	1992.47	1409.57	1681.84	151545.65	389.29
Internal Bond (N/mm ²)	3.53	3.39	1.33	7.70	6.37	3.35	2.69	1.64
SWT- Surface (N)	347.91	325.33	128.63	588.96	460.33	353.37	13105.07	114.48
SWT- Edge (N)	200.75	182.11	85.98	490.04	404.06	197.05	8168.86	90.38

The data were analyzed statistically with one-way factor ANOVA, Tukey HSD test, and regression analysis. There can be seen a positive correlation between the MOR/MOE and the density. The linear regression between the MOR and the density (Figure 51) shows a small statistically significant dependence ($R = 0.5074$). As seen in Figure 52 the correlation coefficient for the MOE and the density was 0.8149. The higher the density the higher the modulus of elasticity. No other correlations were observed in the analysis (Sobotková, et al., 2021).

There are also described the ANOVA results at Figure 53 and Figure 54 which are showing that the CB had a higher MOR and internal bond value than the OP. It was observed that material A (CB) had higher values than material B (OP). The reasoning behind this could possibly be caused by better mixed substances in the mixture and higher strength of cardboard in comparison with regular office paper as discussed on the previous pages. The Tukey HSD values for the MOE and the internal bond were 0.025 and 0.0001, respectively. It can be concluded that there was a statistically significant difference between material A (CB) which had higher values than material B (OP) according to Tukey test HSD for MOE and internal bond. The MOE of CB was higher than MOE of OP and, in case of the internal bond there was a statistically significant difference - material A (CB) had higher internal bond than material B (OP) (Sobotková, et al., 2021).

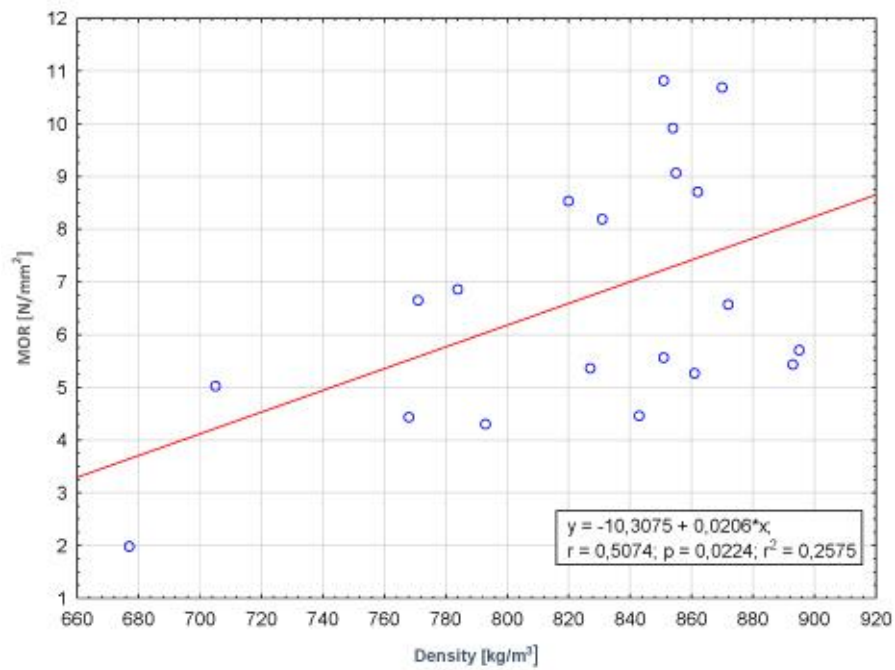


Figure 51 Linear regression between Modulus of rupture (MOR) and density

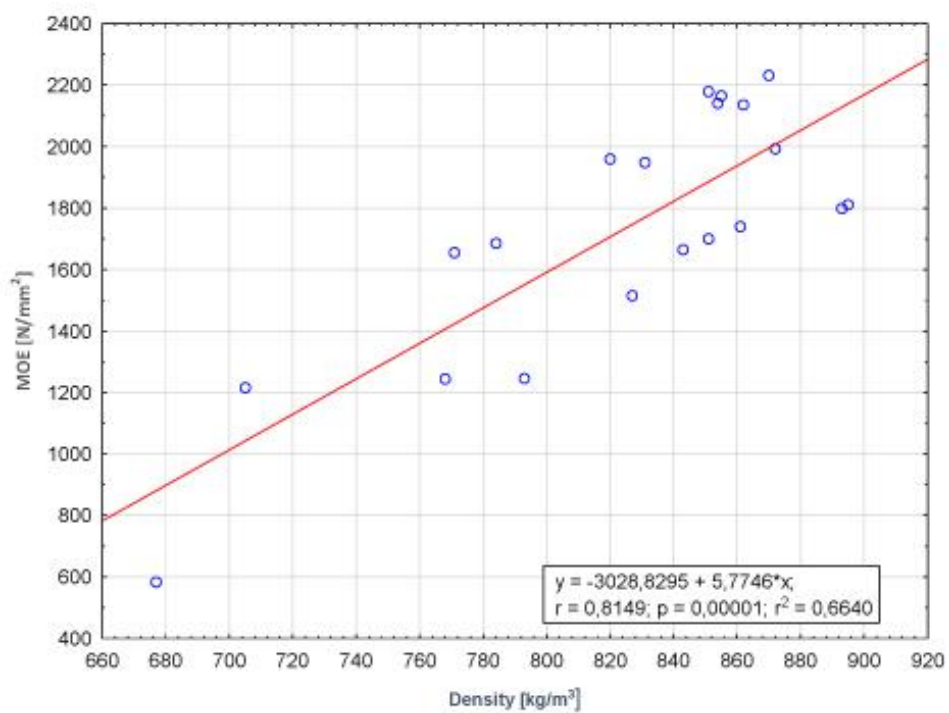


Figure 52 Linear regression between Modulus of elasticity (MOE) and density

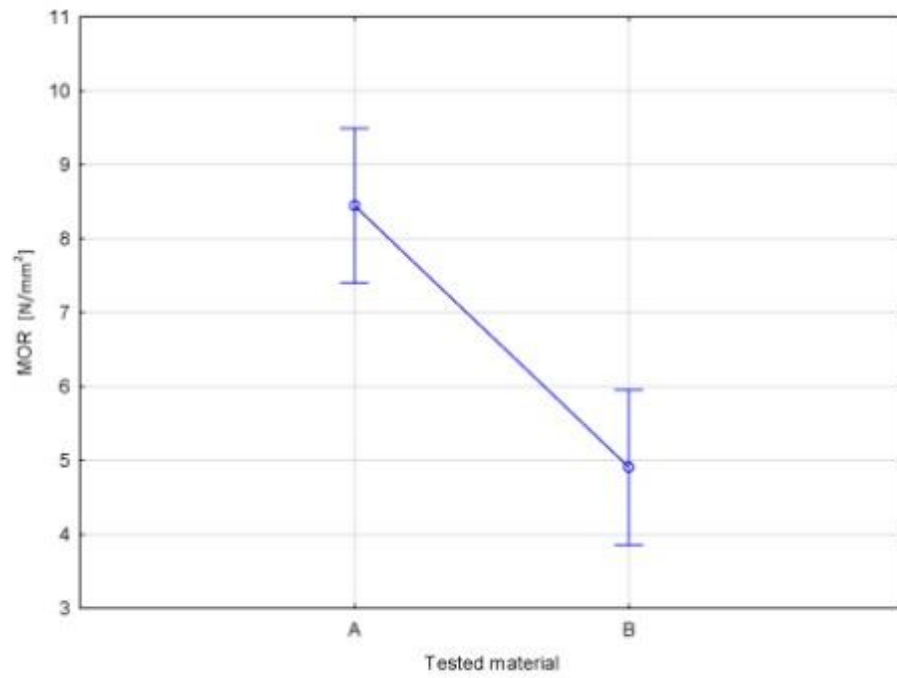


Figure 53 One-way factor ANOVA for Modulus of Rupture and tested materials

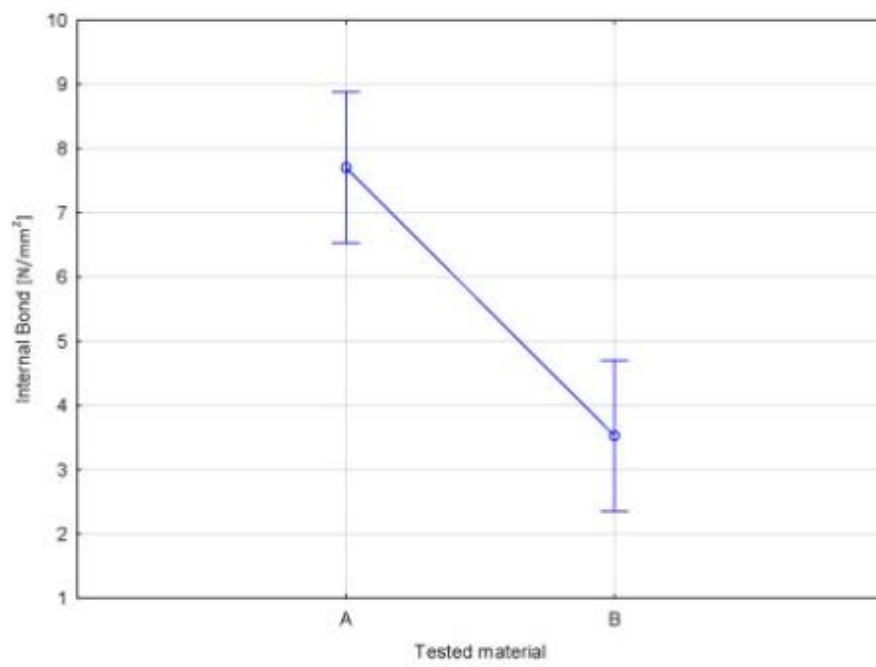


Figure 54 One-way factor ANOVA for Internal bond and tested materials

5.2.3 Results of testing VOCs

The testing of VOCs was done only in the case of paper-PLA boards, since they had significantly better results than the previous manufactured boards from DIP and OCC with the starch or UF glues. The testing of VOCs for both CB and OP boards was done after two months from the manufacturing of the boards. The first measuring was done for the CB board and continued with the measuring of OP boards. The temperature of the environment was 23 °C and the relative humidity was 50 % during the measuring. FLEC methods were used for the characterization of TVOCs emitted from the manufactured paper-PLA boards. According to the results in the Table 17 we can see that total of 31 compounds was measured by the FLEC testing method. Most of the VOCs which were found with the amount higher than 2 µg/m³ were the compounds which can be regularly found in the wood or needles from the trees (hexanal, limonene etc. – see 3.10.1).

5.2.3.1 CB Ø 450 mm, thickness 18 mm

The CB with the diameter of 450 mm was tested. In the Figure 55 the graph shows the compounds which were emitted the most and they are higher than 2 µg/m³ after 72 hours and after 28 days. The most emitted compound was n-butyl acetate with the value of 7.6 µg/m³. The second most emitted was hexanal with the value of 2.8 µg/m³.

Other compounds had negligible portion in the testing sample since they were less than 2 µg/m³ as presented in the Figure 56. Eight compounds were identified the least and their presence was less than 0.1 µg/m³ (Trichloroethylene, Tetrachloroethylene, 4-Ethyl-Toluen, 1,3,5-Trimethyl-Benzene, 2-Ethyl Toluene, Myrcene, 1,2,4-Trimethyl-Benzene, α-Phellandrene).

The TVOCs of the CB 450 18 mm was 24 µg/m³ after 72 hours. After 28 days it was even less – 14 µg/m³.

Table 17 VOC of CB, thickness 18 mm

Number	Compound	After 72 hours CB 18 mm	After 28 days CB 18 mm
	Unit	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
1	Ethyl acetate	(1,5 $\pm$ 0,5)	(0,1 $\pm$ 0,03)
2	Benzene	(0,1 $\pm$ 0,03)	(0,2 $\pm$ 0,1)
3	1-Methoxy-2-Propanol	(0,3 $\pm$ 0,1)	< 0,1
4	Pentanal	(1,2 $\pm$ 0,4)	(0,5 $\pm$ 0,2)
5	Trichloroethylene	< 0,1	< 0,1
6	Toluene	(0,7 $\pm$ 0,2)	(1 $\pm$ 0,3)
7	Hexanal	(2,8 $\pm$ 0,8)	(1 $\pm$ 0,3)
8	Tetrachloroethylene	< 0,1	(0,1 $\pm$ 0,03)
9	n-Butyl acetate	(7,6 $\pm$ 2,3)	(1,1 $\pm$ 0,3)
10	Ethylbenzene	(0,5 $\pm$ 0,2)	(0,4 $\pm$ 0,1)
11	m,p-Xylen	(1,7 $\pm$ 0,5)	(0,9 $\pm$ 0,3)
12	Styrene	(0,1 $\pm$ 0,03)	(0,1 $\pm$ 0,03)
13	o-Xylen	(0,7 $\pm$ 0,2)	(0,3 $\pm$ 0,1)
14	Butoxy-Ethanol	(0,2 $\pm$ 0,1)	(0,1 $\pm$ 0,03)
15	α -Pinen	(1,2 $\pm$ 0,4)	(0,1 $\pm$ 0,03)
16	Camphene	(0,1 $\pm$ 0,03)	< 0,1
17	3-Ethyl-Toluen	(0,1 $\pm$ 0,03)	(0,2 $\pm$ 0,1)
18	4-Ethyl-Toluen	< 0,1	< 0,1
19	1,3,5-Trimethyl-Benzen	< 0,1	< 0,1
20	β -Pinen	(0,3 $\pm$ 0,1)	(0,1 $\pm$ 0,03)
21	2-Ethyl Toluene	(0,1 $\pm$ 0,03)	(0,1 $\pm$ 0,03)
22	Myrcene	< 0,1	< 0,1
23	1,2,4-Trimethyl-Benzen	(0,1 $\pm$ 0,03)	(0,2 $\pm$ 0,1)
24	α -Phellandrene	< 0,1	< 0,1
25	3- δ -Caren	(0,1 $\pm$ 0,03)	< 0,1
26	1,4 dichlorobenzene	< 0,1	< 0,1
27	1,2,3-Trimethyl-Benzen	< 0,1	< 0,1
28	Limonene	(1 $\pm$ 0,3)	(0,8 $\pm$ 0,2)
29	γ -Terpinene	< 0,1	< 0,1
30	Bornyl Acetate	< 0,1	< 0,1
31	TVOCs	(24 $\pm$ 7)	(14 $\pm$ 4)

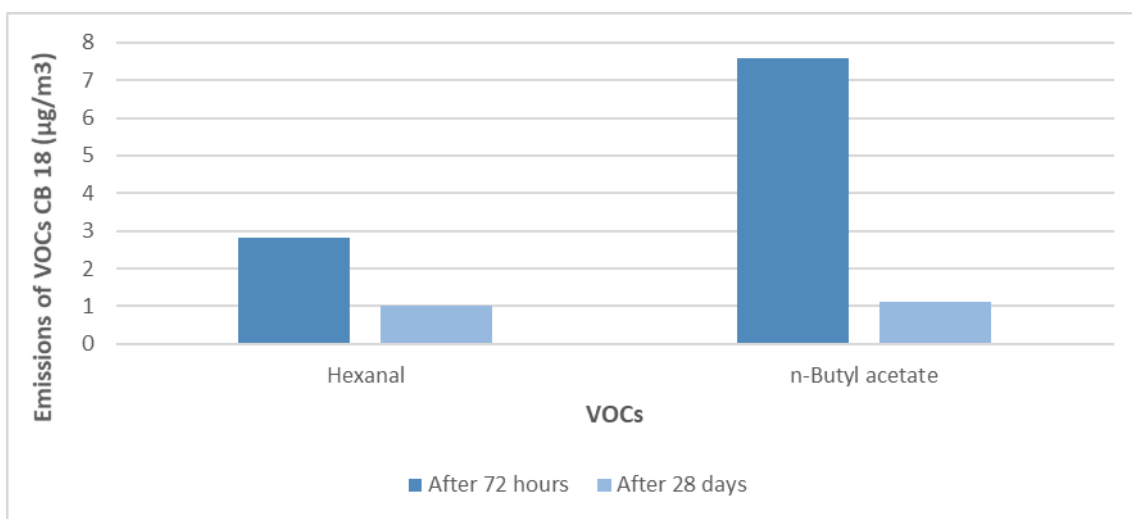


Figure 55 Emission of VOCs >2 µg/m³ for CB 18 mm

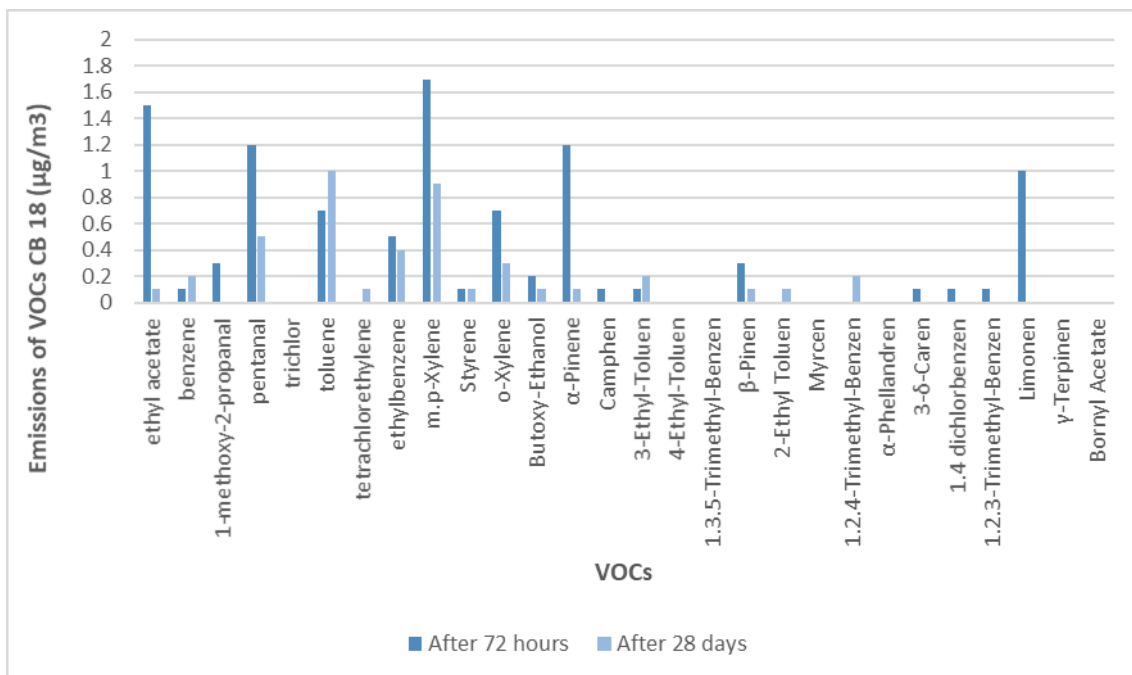


Figure 56 Emission of VOCs <2 µg/m³ for CB 18 mm

5.2.3.2 OP Ø 355 mm, thickness 18 mm

The OP with the diameter of 355 mm was also tested. According to the results in the Table 18 we can see that total of 31 compounds was evaluated during the FLEC test. The most emitted compound can be seen in the Figure 57 in case of OP 18 mm. The biggest portion was limonene with the value of $3.9 \mu\text{g}/\text{m}^3$, $3.3 \mu\text{g}/\text{m}^3$ emissions had hexanal. The third most emitted was n-butyl acetate with the value of $2.3 \mu\text{g}/\text{m}^3$. Other compounds which can be seen in Figure 58 had the value less than $2 \mu\text{g}/\text{m}^3$ and we can state that they take negligible percentage in the testing sample.

The TVOCs of the OP 355 18 mm was $30 \mu\text{g}/\text{m}^3$ after 72 hours. After 28 days it was even less – $12 \mu\text{g}/\text{m}^3$. According to the results compared to MDF boards, where the TVOCs should be less than $200 \mu\text{g}/\text{m}^3$ we can state that the board consists very low number of VOCs and TVOCs and OP fulfills the requirements for such boards.

Table 18 VOC of OP, thickness 18 mm

Number	Compound	After 72 hours OP 18 mm	After 28 days OP 18 mm
	Unit	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
1	Ethyl acetate	$(0,9 \pm 0,3)$	$(0,1 \pm 0,03)$
2	Benzene	$(0,2 \pm 0,1)$	$(0,3 \pm 0,1)$
3	1-Methoxy-2-Propanol	$(0,3 \pm 0,1)$	$< 0,1$
4	Pentanal	$(1,2 \pm 0,4)$	$(0,2 \pm 0,1)$
5	Trichloroethylene	$< 0,1$	$< 0,1$
6	Toluene	$(1,2 \pm 0,4)$	$(0,6 \pm 0,2)$
7	Hexanal	$(3,3 \pm 1)$	$(0,3 \pm 0,1)$
8	Tetrachloroethylene	$(0,1 \pm 0,03)$	$(0,1 \pm 0,03)$
9	n-Butyl acetate	$(2,3 \pm 0,7)$	$(0,2 \pm 0,1)$
10	Ethylbenzene	$(0,5 \pm 0,2)$	$(0,2 \pm 0,1)$
11	m,p-Xylen	$(1,2 \pm 0,4)$	$(0,4 \pm 0,1)$
12	Styrene	$(0,3 \pm 0,1)$	$< 0,1$
13	o-Xylen	$(0,6 \pm 0,2)$	$(0,2 \pm 0,1)$
14	Butoxy-Ethanol	$(0,6 \pm 0,2)$	$(0,2 \pm 0,1)$
15	α -Pinen	$(0,9 \pm 0,3)$	$(0,1 \pm 0,03)$
16	Camphene	$(0,1 \pm 0,03)$	$(0,1 \pm 0,03)$
17	3-Ethyl-Toluen	$(0,2 \pm 0,1)$	$< 0,1$
18	4-Ethyl-Toluen	$< 0,1$	$< 0,1$
19	1,3,5-Trimethyl-Benzen	$(0,1 \pm 0,03)$	$< 0,1$
20	β -Pinen	$(0,4 \pm 0,1)$	$(0,1 \pm 0,03)$
21	2-Ethyl Toluene	$(0,1 \pm 0,03)$	$< 0,1$
22	Myrcene	$< 0,1$	$< 0,1$
23	1,2,4-Trimethyl-Benzen	$(0,3 \pm 0,1)$	$(0,2 \pm 0,1)$
24	α -Phellandrene	$< 0,1$	$< 0,1$
25	3- δ -Caren	$(0,3 \pm 0,1)$	$(0,1 \pm 0,03)$
26	1,4 dichlorobenzene	$< 0,1$	$< 0,1$
27	1,2,3-Trimethyl-Benzen	$(0,1 \pm 0,03)$	$< 0,1$
28	Limonen	$(3,9 \pm 1,2)$	$(1 \pm 0,3)$
29	γ -Terpinen	$< 0,1$	$< 0,1$
30	Bornyl Acetate	$< 0,1$	$< 0,1$
31	TVOCs	(30 ± 9)	(12 ± 3)

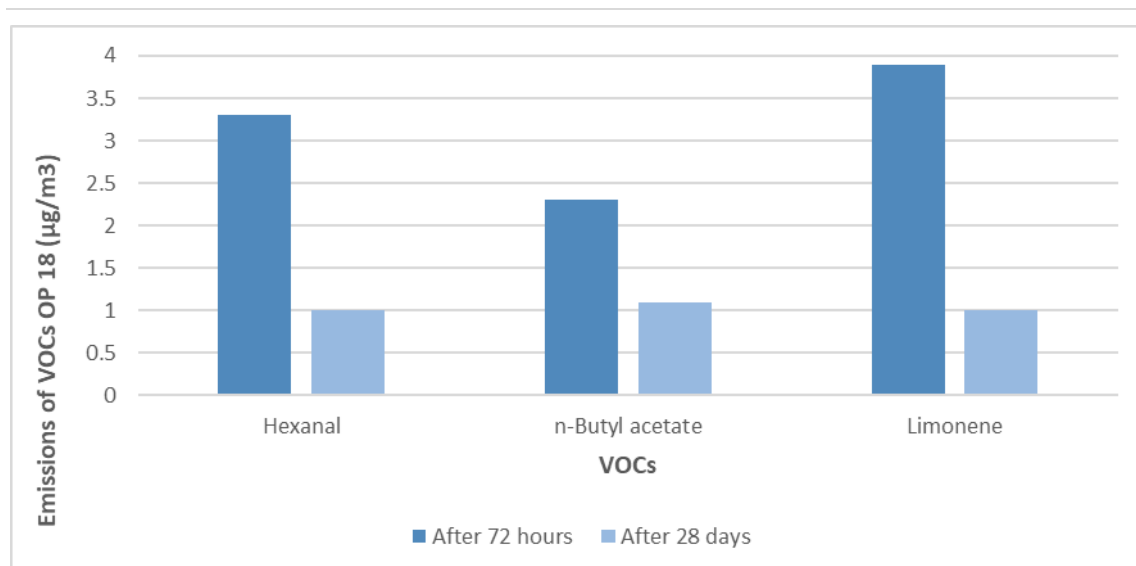


Figure 57 Emissions of VOCs $>2 \mu\text{g}/\text{m}^3$ for OP 18 mm

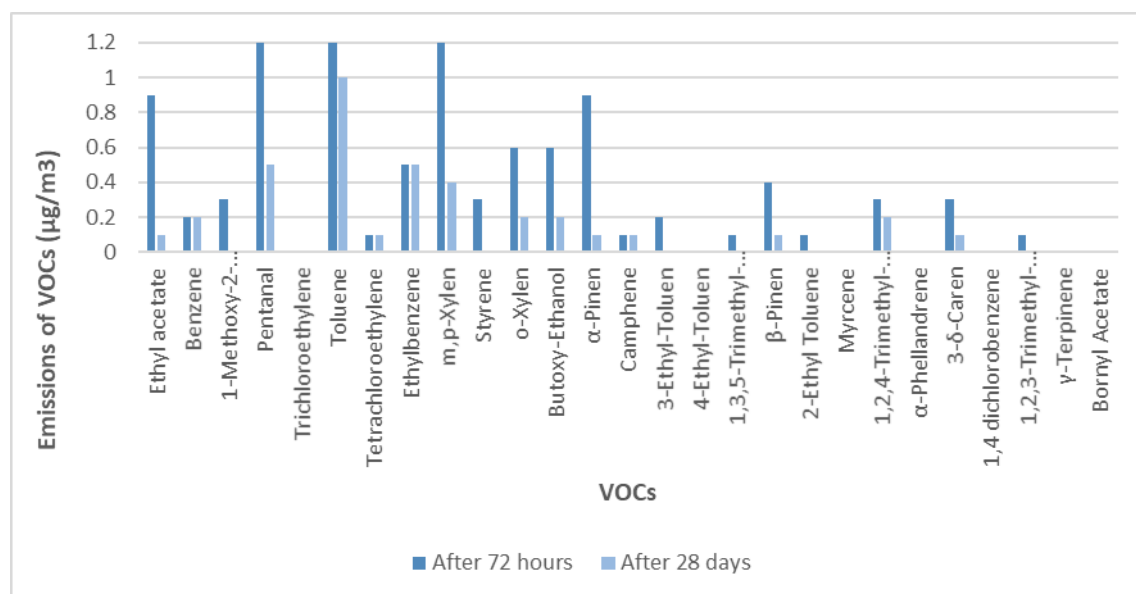


Figure 58 Emissions of VOCs $<2 \mu\text{g}/\text{m}^3$ for OP 18 mm

5.2.3.3 CB 500 x 500 mm, thickness 8 mm

The CB with the dimensions of 500 x 500 mm was also evaluated. According to the results in the Table 19 we can see that total of 31 compounds was found inside of the board. In the Figure 59 it is visible that the highest number of VOCs had limonene (2.6 $\mu\text{g}/\text{m}^3$) and hexanal (2.1 $\mu\text{g}/\text{m}^3$) after 72 hours. In the graphs it is obvious that after 28 days, it is much less. Only in case of m,p-Xylen it was more after 28 days – 1.3 $\mu\text{g}/\text{m}^3$ comparing to 0.5 $\mu\text{g}/\text{m}^3$ after 72 hours but this results is still negligible because it is less than >2 $\mu\text{g}/\text{m}^3$. In the Figure 60 we can also see the negligible compounds which were detected in the testing sample.

The TVOCs of the CB 500 x 500 mm 8 mm was 32 $\mu\text{g}/\text{m}^3$ after 72 hours. After 28 days it was even less – 15 $\mu\text{g}/\text{m}^3$.

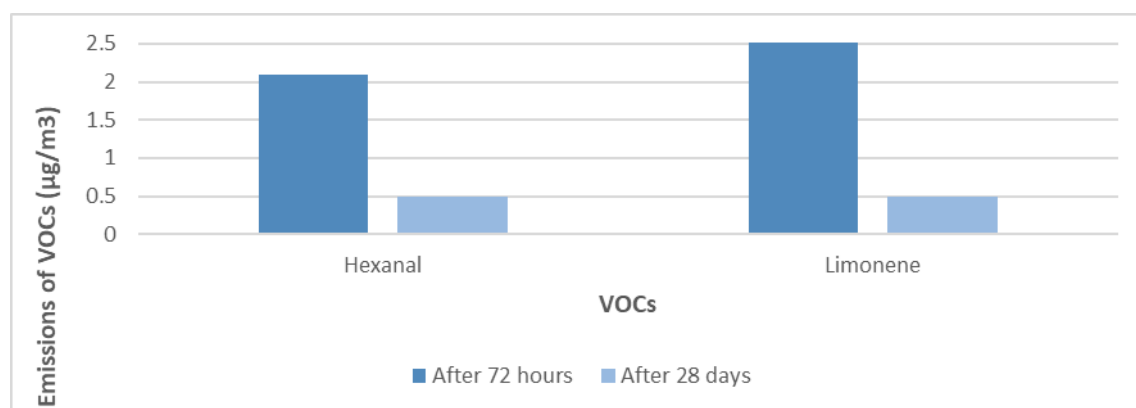


Figure 59 Emissions of VOCs >2 $\mu\text{g}/\text{m}^3$ for CB 8 mm

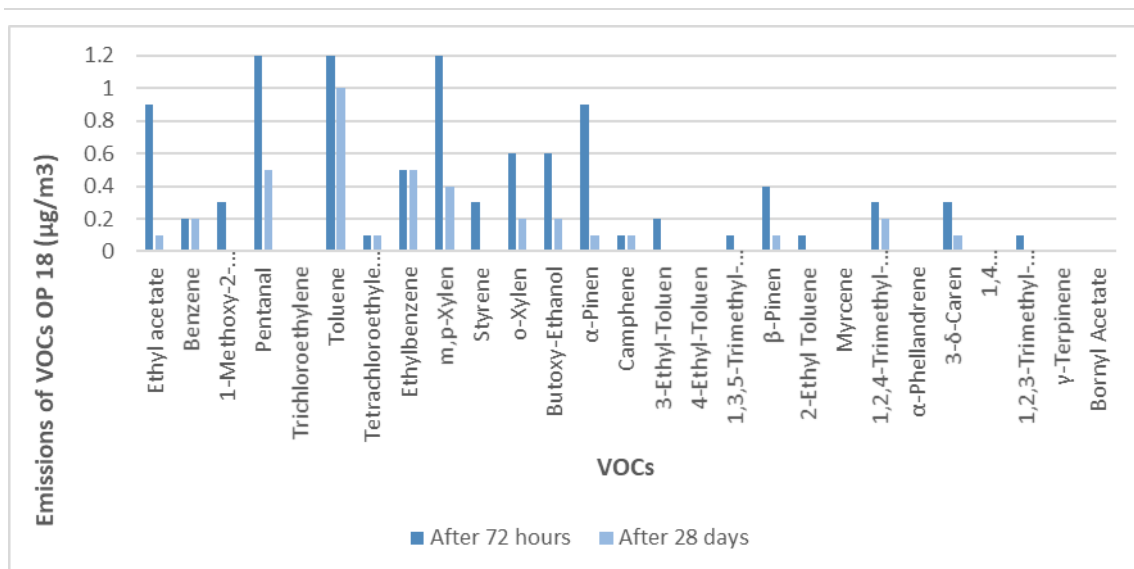


Figure 60 Emissions of VOCs <2 µg/m³ for CB 8 mm

Table 19 VOC of CB, thickness 8 mm

Number	Compound	After 72 hours OP 18 mm	After 28 days OP 18 mm
	Unit	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
1	Ethyl acetate	$(0,1 \pm 0,03)$	$(0,2 \pm 0,1)$
2	Benzene	$(0,2 \pm 0,1)$	$(0,3 \pm 0,1)$
3	1-Methoxy-2-Propanol	$(0,5 \pm 0,2)$	$(0,1 \pm 0,03)$
4	Pentanal	$(1,2 \pm 0,4)$	$(0,2 \pm 0,1)$
5	Trichloroethylene	$< 0,1$	$< 0,1$
6	Toluene	$(0,5 \pm 0,2)$	$(1 \pm 0,3)$
7	Hexanal	$(2,1 \pm 0,6)$	$(0,5 \pm 0,2)$
8	Tetrachloroethylene	$(0,1 \pm 0,03)$	$< 0,1$
9	n-Butyl acetate	$(0,1 \pm 0,03)$	$(0,1 \pm 0,03)$
10	Ethylbenzene	$(0,2 \pm 0,1)$	$(0,3 \pm 0,1)$
11	m,p-Xylen	$(0,5 \pm 0,2)$	$(1,3 \pm 0,4)$
12	Styrene	$(0,1 \pm 0,03)$	$< 0,1$
13	o-Xylen	$(0,2 \pm 0,1)$	$(0,5 \pm 0,2)$
14	Butoxy-Ethanol	$(0,2 \pm 0,1)$	$< 0,1$
15	α -Pinen	$(0,2 \pm 0,1)$	$(0,1 \pm 0,03)$
16	Camphene	$< 0,1$	$(0,1 \pm 0,03)$
17	3-Ethyl-Toluen	$(0,2 \pm 0,1)$	$(0,1 \pm 0,03)$
18	4-Ethyl-Toluen	$(0,1 \pm 0,03)$	$(0,1 \pm 0,03)$
19	1,3,5-Trimethyl-Benzen	$< 0,1$	$< 0,1$
20	β -Pinen	$(0,1 \pm 0,03)$	$< 0,1$
21	2-Ethyl Toluene	$(0,1 \pm 0,03)$	$< 0,1$
22	Myrcene	$< 0,1$	$< 0,1$
23	1,2,4-Trimethyl-Benzen	$(0,2 \pm 0,1)$	$(0,4 \pm 0,1)$
24	α -Phellandrene	$< 0,1$	$< 0,1$
25	3- δ -Caren	$(0,1 \pm 0,03)$	$(0,1 \pm 0,03)$
26	1,4 dichlorobenzene	$< 0,1$	$< 0,1$
27	1,2,3-Trimethyl-Benzen	$(0,1 \pm 0,03)$	$< 0,1$
28	Limonene	$(2,6 \pm 0,8)$	$(0,5 \pm 0,2)$
29	γ -Terpinene	$< 0,1$	$< 0,1$
30	Bornyl Acetate	$< 0,1$	$< 0,1$
31	TVOCs	(32 ± 10)	(15 ± 4)

Finally, the TVOCs of all the results were compared. According to Table 20 and can observe that the highest amount of TVOC had the CB 8 mm thick board with the dimensions of 50 x 50 mm. After 72 hours it was 32 $\mu\text{g}/\text{m}^3$ and 15 $\mu\text{g}/\text{m}^3$ after 28 days. Comparing to industrially manufactured boards such as MDFs, the value fulfills the requirements which are less than 200 $\mu\text{g}/\text{m}^3$ according to the EN 16516.

Table 20 Comparison of TVOC of CB and OP boards with MDF boards (EGGER)

UNIT	TVOC 72 H	TVOC 28 DAYS
	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
CB 450 18	24	14
OP 350 18	30	12
CB 8 MM 50 X 50	32	15
EGGER MDF (ČSN EN 16516)	-	<200

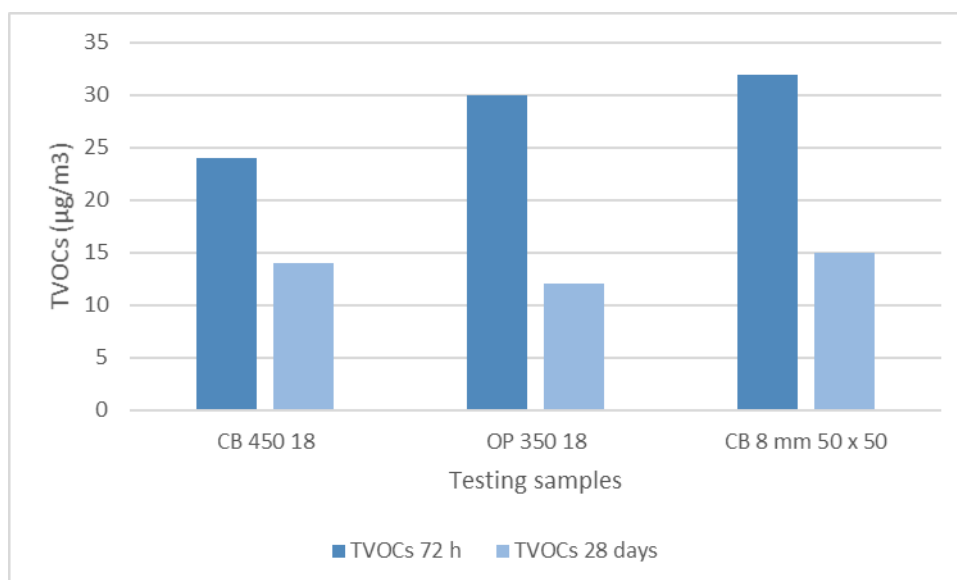


Figure 61 TVOCs of testing samples

5.2.4 Discussion – paper-PLA boards

The properties of both CB and OP had very interesting results in terms of their use for wood-based products and furniture. These boards can be compared with boards for interior fitments (including furniture) for use in dry conditions (Type P2) and non-load bearing boards for use in humid conditions (Type P3). They have an anisotropic behavior because of the fiber orientation in paper.

The properties of both materials were dependent on how many times the paper was milled together with PLA and limestone. CB was milled four times unlike OP which was milled only two times. Also, the size of OP fibers is much smaller than CB fibers.

The data was analyzed statistically through regression analysis and one-way factor ANOVA and the Tukey HSD test. There was a positive correlation between MOR/MOE and density. The linear regression between modulus of rupture and density which can be seen at

Figure 51 shows that there is a small statistically significant dependence ($r = 0,5074$). Correlation coefficient for modulus of elasticity and density is 0,8149 (Figure 52) which means that there is a statistically significant dependence between these two variables. The higher density the higher modulus of elasticity.

Another statistical processing of data was one-way factor ANOVA. It was made for both materials and each testing method. According to the Tukey test HSD there is a statistically significant difference between material A and B. As Figure 53 shows, material A (CB) has higher MOR than material B OP. At Figure 54 shows ANOVA for internal bond. There was a statistically significant difference between both materials and material A (CB) had higher values than material B (OP).

Both CB and OP paper-PLA boards had better results than the DIP and OCC boards, which is caused by added plastic and limestone powder. Also, the dry pressing can be considered as more suitable for such a process. The

testing of VOC confirmed that none of the manufactured paper-plastic boards from CB and OP are emitting harmful substances, which can negatively affect the users of the designed furniture which should be used in interior.

Overall CB had the best physical and mechanical properties than OP boards and the paper-pulp boards. The reason is caused by the natural strength of CB in comparison with regular office paper. This testing has brought a result that the CB material can be used to produce paper-plastic boards as other possibility of the manufacture of wood-based products such as particleboards or MDF boards for a small tabletop. The boards were also manufactured in a sustainable and circular way, because they can be broken down into smaller parts, milled and the new boards can be manufactured.

Thanks to the evaluation of the VOCs and TVOCs emitted from the boards we can also say that the manufactured boards are non-harmful to the environment and the user. In the Chapter 0 we can see that the highest amount of TVOCs had the 8 mm CB boards where the value was $32 \mu\text{g}/\text{m}^3$ after 72 hours and $15 \mu\text{g}/\text{m}^3$ after 28 days. Comparing to industrially manufactured boards it is much less than the standard value which should be less than $200 \mu\text{g}/\text{m}^3$. Thanks to this experimental part we can confirm that the boards should be non-harmful to the users, and they can be used in the interiors.

6. Applications of the results

The research started with the idea of reusing waste material and using it for sustainable furniture. The main goal was to design furniture with the application of the sustainable material to show how the materials could be used. It was not designed to substitute traditional materials, which are already used but the idea was to show other options in furniture design and application of sustainable or upcycled materials.

The first design was done during a one-year project sponsored by Internal Grant Agency at Mendel University in Brno, Faculty of Forestry and Wood Technology. The design process started with market research which can be seen in chapter 3. The biggest inspiration for the final product was a project of the Canadian company Dear Human and their paper pulp furniture. The research started with the testing of paper pulp and its physical and mechanical properties. Later other combinations of materials were tested. After the evaluation of all the materials which were manufactured and their physical and mechanical properties were tested, the best option was chosen for the final product. The boards from CB and OP with PLA and limestone were manufactured. These boards were used for the design of coffee tables PapeerStack.

Tables were used as a main object also for the biggest design event in the Czech Republic – International Design Festival Designblok 2021 in Prague, which was the first-time since the Mendel University in Brno exhibited at this big design event. The author of the thesis designed the whole installation of the university stand.

6.1 Coffee tables PapeerStack

The first products which were made with the paper-PLA boards were stackable tables, PapeerStack. The top desk is made from CB and OP and the boards where the binder was biodegradable plastic PLA. The development of

the design started with research of the market and the products which are being sold. The author of the thesis started with the ideation process to develop the right shape. The first sketches were made. When the right shape was found, the best idea was chosen, and the 3D visualization was done. It continued with the technical drawing for the manufacturer of the tables. Finally, the coffee tables were manufactured. The whole process will be described at the following pages.

6.1.1 Sketches

Sketching is an important tool for every designer. First ideas are drawn to find the right shapes. The main goal was to design a table with the idea of RTA (Ready-to-assemble) furniture. The most important was to think about the joint of the paper composite board with the wooden construction. There have been different ideas and design until the most satisfying solution was found. The process of work and the final design can be seen at the pictures below (Figure 62&Figure 63).

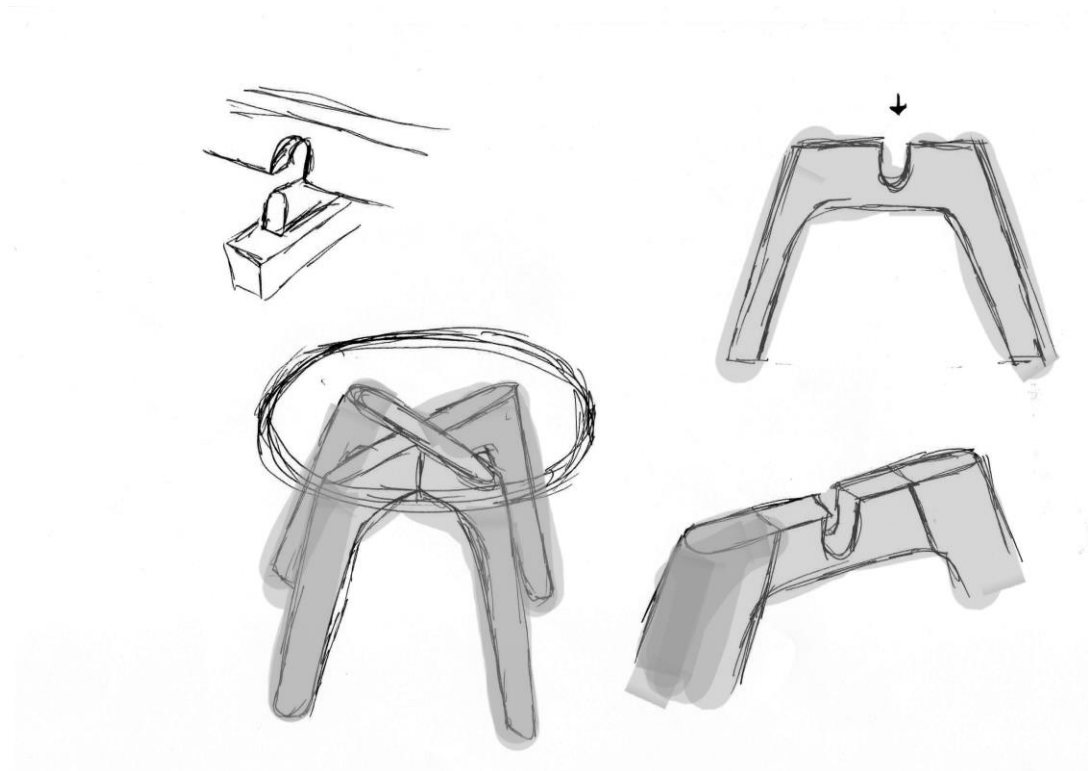


Figure 62 Looking for a shape of tables



Figure 63 Final shape of the tables

6.1.2 Construction

The construction of the tables is very simple. It consists of two wooden parts and the paper desk which is joined with only one screw. The main goal was to design a simple coffee table with the principles of RTA, so it is very easy to disassemble the tables and move them.

The dimensions of tables are following the ergonomic and anthropological dimensions which are suitable for the use of people. The height of the higher table is 500 mm and the smaller is 450 mm. The table with CB board has a diameter of 450 mm and OP has a diameter of 350 mm. Technical drawing for the manufacturing of the tables can be seen in the appendices (Appx. 4 – appx. 7).

6.1.3 Material

The tables are made from two materials. The used material for the top desks of the tables were paper-PLA boards, which were manufactured at

Mendel University in Brno, Department of Wood Science and Technology in Útěchov as part of the IGA project in 2018/2019. The two types of used paper were old, corrugated cardboards and office paper. Top desks can have different look and the final color and structure is depending on the used type of paper (Figure 64). For the construction of the tables was used oak wood. Oak is a resistant wood with density about 750 kg/m^3 . Other types of wood could be used according to the needs of the customer.



Figure 64 Paper boards with PLA (author's archive, 2020)

6.1.4 Visualization

Before the final design was created, the 3D model of the tables was done. The 3D model is crucial for knowing if the final design would work. It is great tool how to see how it will look like also in the interior, and if the joints are suitable for the design solution. It also helped to imagine the product in the interior. The visualizations can be seen in the Figure 65 and Figure 66.



Figure 65 Visualization of the set of tables PapeerStack



Figure 66 Vizualization of two coffee tables PapeerStack

6.1.5 Final design

Final design of the tables was designed with the use of paper-plastic boards as a top desk for the table (Figure 67). Final design of the tables can be seen at pictures below (Figure 68) and in the appendices (Appx. 8). The value of the product is in its aesthetics thanks to different surface of the paper which has stone-like look. Oak wood is durable and has long durability. Tables are supposed to be used in interior and they can be stacked on each other (Figure 72). Surface treatment is considered for other products.

The design was supposed to be simple, and it should correspond with the lifestyle of young people who are traveling quite often. That is why it is very easy to disassemble with only one screw (Figure 69&Figure 71).

Final look of the top desk from paper can be also influenced by the chosen type of paper. Different paper can lead to different looks of the surface as seen on the samples below. The legs of the table can be manufactured from different type of wood.



Figure 67 Surface looks (author's archive, 2020)



Figure 68 Coffee Tables CB and OB with oak wood (Zvěřina Photography, 2020)



Figure 69 Coffee Tables dissassembled (Zvěřina Photography, 2020)



Figure 70 PapeerStack collection of tables (AVC MENDELU, 2021)



Figure 71 PapeerStack – detail of the joint (AVC MENDELU, 2021)



Figure 72 PapeerStack collection of stackable tables (AVC MENDELU, 2021)

6.2 Design of exhibition at Designblok 2020

During the research, the author started to cooperate with a company Plastia, s.r.o. who was interested in using paper-based material for their products. The aim was to show the people that there are also other options while creating new products. It should represent the possibility of the use of recycled paper together with post-factory and post-consumer plastic materials at their plastic flowerpots and to spread the ecological thinking between the visitors.

During several meetings with the head of the company, the cooperation was established for an exhibition at the biggest International Design Festival in the Czech Republic – Designblok 2020 in Prague. Together with the company Plastia the author designed and helped with the installation of the whole exposition at Designblok 2020. The author of the thesis designed the whole exposition, made technical drawings and visualizations of the space. She also designed info graphics and manufactured with laser technology at the Department of Furniture, Design and Habitation.

The goal of this exposition was to show the idea of recycling materials together with flowerpots of the company which would be exhibited with plants inside. That is why it was named by eco-jungle. Main objects of the expositions should be the project of paperboards, recycled material from the company which was used during the workshop *“Vratte je do hry”* and also flowerpots made from post-factory and postconsumer plastics.

More paper boards were supposed to be used for the design of storage systems, although due to the pandemic situation it was possible to develop only two more boards.

Since it was not possible to manufacture more boards and only two boards with the dimensions of 500 x 500 mm were manufactured, the design of the system changed. In the end as a display were used two paper-plastic boards which were hanged at the constructional metal webs. They were used for the design of exposition at Designblok 2020 which was done together with

the company Plastia.

The whole process of the preparations for the exhibition started during January 2020, while it was necessary to talk about the cooperation between the university and the company and decide about the concept of the exhibition for the topic which was Passion in 2020. The application for the exhibition was due to May 2020 when the final visualization had to be done together with the description of the show. In June 2020, the author got the message from the organizers of Designblok 2020 that the exposition was chosen for the festival. The preparation process has started.

6.2.1 Technical drawings

Technical drawing was done for the company who installed the whole exposition. The technical drawings (Figure 73&Figure 74) can be also seen printed as the appendices of this work (Appx. 9 - appx.12). The center of the room was given to the coffee tables and the lighting “Medůza” which was hanged above the tables. Behind the tables there was supposed to be an info panel with key words like re-use, re-duce, and re-cycle.

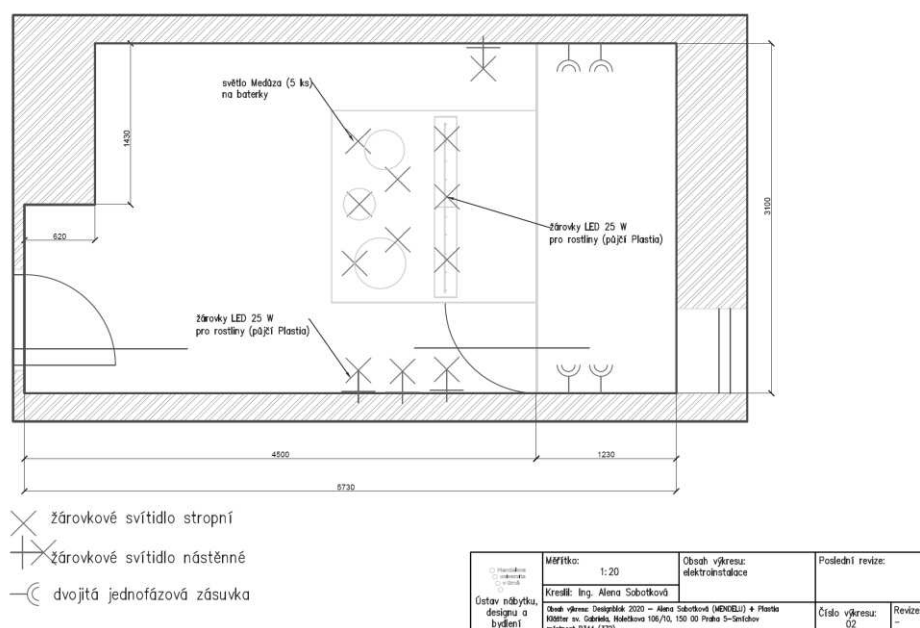


Figure 73 Electroinstalation (appx. 10)

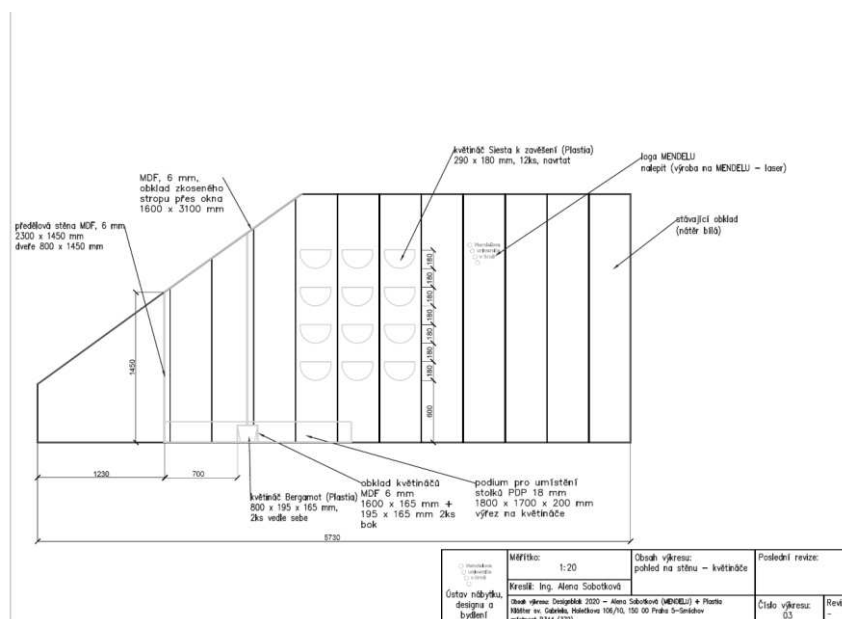
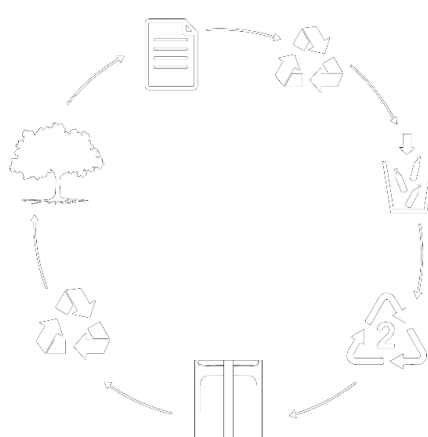


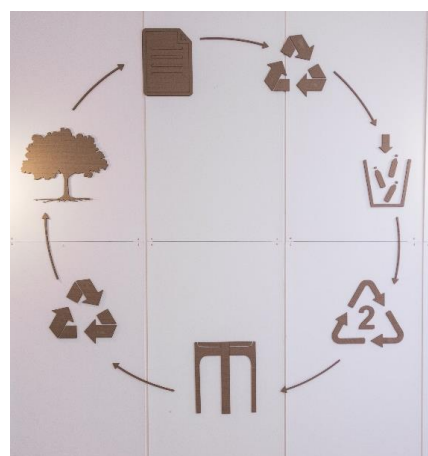
Figure 74 View on the wall (appx. 11)

6.2.2 Laser cutting

Part of the exposition was an infographic to represent the idea of reusing materials. First, there is a tree, which means wood. Wood is the main substance for the manufacturing of paper. Paper can be recycled then which is the same as in the case of plastic. After these materials are recycled, they can be re-used for the manufacturing of paper-based boards. The author designed this circular wheel to present the idea to the visitors (Figure 75).



a.



b.

Figure 75 a. – Data for laser cutting; b. - Laser cutted wall decoration of cardboard

6.2.3 Visualization of the exposition

After many consultations with the company Plastia s.r.o. and with the author's supervisor and the final design of exposition was chosen (Figure 76). The main idea was to show different projects from students of MENDELU, who are working on sustainable design projects. The main goal was to show the paper boards used for furniture design.

The center of the exposition was assigned to the PapeerStack coffee tables as the main piece. Behind the tables should be an info panel with the key words such as re-duce, re-use, re-cycle and with the name of the exposition itself – eco jungle. On the walls there was an infographic with the aim of showing how the process of recycling and upcycling works. On the other wall there was a place for flowerpots from the company Plastia s.r.o. made from post-consumer plastic. Another exhibited student project was the lights "Medusa" which were designed by Michela Kubátová, a student from MENDELU, who was cooperating with the company Plastia as well.



Figure 76 Visualization of the exhibition at Designblok 2020

6.2.4 Final exposition

The design of the exposition went through a few changes before the realization. First, there was not enough material available for pressing more boards, so the info panel was made from MDF. Two paper boards were manufactured in the end. The paper boards were hanged on the constructional webs on the walls next to the infographic. The tables were exhibited in the middle of the room. Some plants were exhibited in the idea of “eco jungle” to show the relationship between paper, nature but also plastic and the circular lifetime of each part. The whole idea derives from the rules of circular economy, which is the one of the most important rules in eco-friendly design. The final exposition which was installed can be seen at Figure 77 and more to see in the appendices (Appx. 13).



Figure 77 Exposition at Designblok 2020 (AVC MENDELU, 2020)

7. Future ideas for the research

The author would like to continue with the development of paper based composite boards, which could be used for furniture design, but also other applications such as wall coverings which could serve as an acoustic panel. They could possibly be used as a building material. There are still many options for the use of paper in furniture design. The combinations of paper and plastic is very good for furniture use, since the testing resulted in a very strong board. In comparison with paper-PLA boards, it would be very eco-friendly with only starch adhesive and in a way of reusing or recycling this material after its lifetime ends. While using paper-plastic boards there are crucial steps which must stay such as high-speed milling of the material together with the plastic which is not so easy to do without the right equipment. Unfortunately, the author has not been able to find another company yet, which would have the right equipment and the one, which cooperated before did not have the capacity for further research.

After the exhibition, the author of the thesis received very valuable feedback on her work. Overall, the boards were a great success and people were interested in the idea of recycling. Several questions were asked. The questions will be asked and answered in the following pages:

1) Would it be possible to recycle or reuse this material after its lifetime ends?

Theoretically, it should be possible to mill the boards again into small particles, add more paper and plastic and press the boards again, since there is no glue added. Further development of this problematic and testing this option should be applied.

2) Is there any surface finish used?

No, at the exhibition there was no finishing used, to keep the paper appearance to the boards, but in future development the author tested finishes such as Bluefin Pigmores 4in1 G50 W30 and Bluefin Isospeed which are water-based

and quick-drying covering polyurethane fillers (commonly used for MDF board surfaces). Other surface finishes should be tested afterwards.

3) What types of paper can be used for pressing the boards?

The paper which can be used, should not have a surface finish such as chalk paper have. It should be a sorted paper such as regular office paper. The used paper is affecting the appearance of future boards and it can create different colors but also surfaces.

4) Can other plastic material be used for pressing the boards?

Yes, other granulate plastic can be used. The author started with testing of HDPE (High Density Polyethylene). Also recycled granulate from HDPE could be used. Unfortunately, the further testing, which was supposed to be done at Salzburg University of Applied Sciences during the project AKTION was interrupted by pandemic of COVID-19.

That is why the author continued with testing of previous boards from PLA and she tested VOC and the surface finish.

The whole research can be an example of different use of recycled materials, especially paper. Further development is being discussed with company Plastia but also the author is still looking for other options of its use, since the material could be great supplement to other industrially manufactured materials.

For further development of this topic, it is important to find the right surface finish for the boards which was being pre-tested during the finishing of the thesis, but the results were not ready to be presented.

8. Discussion

This thesis and research are giving a look into the problematic of sustainability, ecology, and circularity in furniture design. The thesis is also trying to increase the awareness about the ecological and sustainable designs of new products for the designers, research new materials which are being created from waste and bio-based materials and the use of waste materials from the manufacturing process of several different companies in student's projects.

The focus was given to the use of wastepaper in combination with adhesives (starch, UF) and biodegradable binder (PLA). Further research stayed with the idea of using paper as a main material for wood-based products and furniture design and pressing of the boards in a same way as the manufacturing process of particleboards or MDF boards was used.

The research came into the process of developing composite material – paper board which can be used as a material for furniture design. Several testing samples were manufactured and tested according to European standards, such as internal bond, modulus of elasticity and modulus of rupture, density profile, screw withdrawal resistance and swelling in thickness and water absorption. The main goal was to find the best binder for paper which would also give it strength and resistance in the use as a constructional material for wood-based products. Several different studies were made and followed during this research.

First trials started with the use of paper pulp (DIP and OCC) during a three-month internship at Salzburg University of Applied Sciences. It started with the manufacturing of paper boards without any additives and 20 mm thickness with different temperatures – 60°C, 120 °C and 180 °C. Next step was to add and adhesive. The author has researched different options of wood adhesives (Wood Handbook, 2010). The most used adhesives in wood-based product industry are the thermosetting resins, which are synthetic polymers. The author wanted to compare the industrially used adhesive with the natural

adhesive, which would be more eco-friendly. The author decided together with her supervisor to use starch adhesive which is often used for paper making and as a glue for wallpapers (4.1.2.1). The pressing temperature was 180 °C. The pressing time was depending on the thickness of the board – in this case it was 15 minutes. Second used adhesive used as a binder was urea-formaldehyde glue. The reason was to use industrially widely used glue which is the most common adhesive for wood-based products such as particleboards or MDF boards (Wood Handbook, 2010). The thickness of the board was 7 mm and 20 mm. All these boards were cut into testing samples and their physical and mechanical properties were tested. The best results were showing UF glue 7 mm thick board due to its high density.

Further research continued with the office paper and old corrugated cardboards from waste centers around Brno, Czech Republic. Paper was cut into smaller pieces and milled by high-energy milling machines after that. PLA was added as a binder of the whole mixture together with smaller amount of limestone to give it better hardness. These boards were cut into testing samples and their physical and mechanical properties were tested. All the results were compared between each other but also with industrially manufactured wood-based products.

It was found out that the best results had the old corrugated cardboards together with PLA and limestone. In comparison with the standardized values of the non-load bearing boards for the use in humid conditions (Type P3) both CB and OP had much better thickness swelling after immersion in water for 24 hours. The average value for CB was 3.73 % and 7.64 % for OP. %. According to EN 312 (2010) the thickness swelling is 13 %. The plastic material is a great binder of paper, and it gives the boards water resistance. Limestone is giving it hardness of the surface.

Another important result is MOR and MOE which was also compared with the EN 312 (2010) where the minimum value of MOR is 11 N/mm² and MOE minimum value of 1600 N/mm². CB had the average MOR value 8.45 11 N/mm² and MOE was 1931.32 N/mm². OP had MOR value of 4.91 N/mm² and MOE

1529.22 N/mm².

Internal bond was another mechanical property which was tested. The mean value of internal bond in the case of CB was 7.70 N/mm² and 3.53 N/mm² in the case of OP. Based on the EN 312 (2010), the minimum value of internal bond for boards for interior fitments (including furniture) for use in dry conditions (Type P2) of boards with 18 mm thickness is 0.35 N/mm².

The mean value of screw withdrawal resistance from the face of CB was 316.59 N and 347.91 N for OP. Screw withdrawal resistance from the edge of CB was 170.93 N and 160.57 N of OP. In comparison with industrially manufactured boards (EGGER), where the resistance for 18 mm board is higher than 1080 N, the results were much lower. We can say that the screw withdrawal resistance is not very satisfying. The reason is caused by small fibers of the paper and because they are losing its strength over many millings. Another joint solution was considered during the process of designing the products which could be manufactured out of the paper composite material.

The data was analyzed statistically and regression analysis, one-way factor ANOVA and Tukey test HSD was done to find out the best results.

Also, the VOCs of the paper-PLA boards were tested. It was found out that the amount of the VOCs is very low. The TVOCs of the boards which was around 20 µg/m³ is much less than the standard value which should be less than 200 µg/m³. We can say that they are non-harmful to the people. Further investigation about the usage of the boards and its properties should follow.

According to the results for testing the paper based composite material, the boards were used in furniture design. The author has designed and manufactured coffee tables with the top desk made from paper boards – PapeerStack. The tables were designed with RTA principles, so it is easy to disassemble and transport them. The top board is made from paper-plastic boards. These tables were the main exhibited piece at the exhibition which was designed by the author of this thesis too – Designblok 2020. Another product was the design of info panel for the exposition at Designblok 2020 which was

made together with the company Plastia s.r.o. Designblok 2020 was held in Prague, Czech Republic from the 7th till the 11th of October 2020. The idea was to exhibit flowerpots made from recycled plastic and show another display made from recycled material – paper boards. In the end the panel was made from constructional webs and the paper boards were hanged on them to show the surface of the material. The company was also interested in the use of the paper boards material for their flowerpots. The whole exposition had main topic of sustainability and circularity in design.

9. Conclusion

During the author's research there were several goals for this thesis.

First, the work was supposed to describe the process of designing new products while keeping in mind sustainability and its environmental impact. It should also explain the important key words such as sustainability, recycling, reducing, reusing, and upcycling.

Second, to map the companies which are designing their products in sustainable idea and connect it with the work of students which was done during a workshop in Křtiny. in cooperation with TON company. Other sustainable materials were found such as rice straw boards, Resysta or MycoComposite.

Another goal was to develop furniture with sustainability in mind. The research on the topic started with 3d printing and came to the use of recycled paper for paper-based board material.

After the boards were manufactured, its physical and mechanical properties were tested. This research was dealing with the upcycling of materials which could be used for wood-based products. The main goal was to find alternative solution of the use of recycled paper.

The main conclusions about the paper-based boards are:

1. The CB and OP post-consumer paper can be used to produce paper-PLA boards as another option of wood-based products.
2. The boards were successfully used as a top desk of a small coffee tables for interior use.
3. Cardboard has better properties than regular office paper because of its manufacturing process. All the results of CB provided better results.
4. Upcycling of paper and plastic was used successfully - The boards were manufactured in a sustainable and circular way because they are

reusing salvaged material and a new life is given to the paper.

5. The boards fulfill the requirements for the TVOCs emitted from the boards.
6. The boards should be non-harmful to the environment.

Thanks to this research there were obtained very valuable results for the use of this type of composite material. The manufactured material was used for design of coffee-tables and storage/info panel.

During the research very important impact had studying and researching abroad at both LSU and SUAS. Each country and university gave the author different approach into the problematics of waste. It could be said that each country is the opposite of one and another. In USA there is a huge amount of waste each year and people are not so used to sort out waste materials. On the other hand, Austrian people are good in sorting out the waste, which is also important approach in sustainable lifestyle.

The results of the research in form of paper-plastic boards were used for furniture design of coffee tables PapeerStack and they were successfully exhibited at furniture design exhibition at the biggest Czech design event – International Design Festival Designblok 2020.

10. Shrnutí

Autorka této disertační práce se zabývá využitím odpadového papíru na materiály na bázi dřeva pro design nábytku. Výzkum této problematiky začal podrobnou rešerší na téma ekologie a udržitelnosti se zaměřením na progresivní technologie, a především využití odpadu jako zdroje pro nové produkty. V rešerši bylo objeveno několik různých materiálů na bázi přírodních materiálů, jako je rýže, škrob, houby ale také dřevo a především papír. Inspirací byla firma Dear Human a také výroba WPC desek. V rešerši bylo poukázáno také na problematiku cirkulární ekonomiky a její pravidla, která jde ruku v ruce s LCA analýzou. Důležitou problematikou jsou také pojmy jako je recyklace, upcyklace ale také downcycling. Okrajově se práce zabývá také vlivem VOC látek ve vnitřním prostředí. V práci jsou popsány materiály a technologie, které byly použity pro praktickou část této práce.

Autorka této disertační práce absolvovala během doktorského studia několik zahraničních stáží. Zde se začala zabývat problematikou odpadního papíru, který je ne vždy možné recyklovat a využít pro další účely. Jedná se například o papír používaný jako obalový materiál, kartony na vejíčka a další. S tímto materiálem autorka začala experimentovat během roční pracovní stáže na LSU v Baton Rouge, USA. V rámci ročního pobytu na LSU se doktorandka aktivně podílela na výuce designu nábytku a 3d modelování na College of Art and Design.

Po získání ročního grantu Interní grantové agentury LDF MENDELU na téma „*Trvale udržitelný rozvoj v procesu navrhování nábytku*“ autorka začala experimentovat s lisovanými deskami z odpadního kancelářského papíru a kartonu ve formě papírové buničiny (DIP a OCC). Nejprve se touto problematikou zabývala v rámci pracovní stáže na SUAS v Kuchlu v Rakousku. Zde byly testovány první vzorky papírových desek nejprve bez přidaných pojiv o různých teplotách lisování a následně testy pokračovaly s využitím škrobového a močovino formaldehydového lepidla. Různé testovací vzorky byly slisovány a následně testovány dle příslušných norem (EN 312: 2010; EN

323: 2005; EN 322: 2005; EN 317: 2005; EN 319: 2005; EN 320: 2011; EN 310: 1993). Tyto výsledky byly porovnány s průmyslově vyráběnými deskami MDF od firmy EGGER. Výsledky práce byly publikovány a prezentovány na konferenci ICWSE 2019 v Brašově v Rumunsku (viz příloha).

Na tento výzkum dále navázalo testování papírových desek spolu s granulátem PLA. Zde byl využit odpadní papír, který byl nadrcen a následně namlet pomocí vysokorychlostních mlýnů, což zaručilo důkladné smísení různých materiálů. Díky tomu došlo k vytvoření téměř homogenního materiálu s vlastnostmi podobnými MDF deskám. Tyto desky byly taktéž testovány po vzoru předchozích experimentů se papírovou buničinou, škrobem a UF lepidly. Výsledky tohoto testování byly publikovány v odborném časopise BioResources (viz příloha). Následně byly otestovány také VOC látky, jelikož se jedná o důležitý faktor ovlivňující možnost využití v interiéru.

Vzhledem k ideálním vlastnostem materiálu, vysoké pevnosti a odolnosti vůči vodě byl tento materiál zvolen pro výrobu prototypu série odkládacích stolků PapeerStack. Stolová deska těchto stolků byla vyrobena z papír-PLA desky (karton, kancelářský papír). Podnož stolků byla vyrobena z masivního dubového dřeva. Cílem bylo ověřit funkčnost desek na jednoduchém designu.

Prototyp stolků byl následně stěžejním bodem pro expozici na výstavě v rámci největšího mezinárodního designu festivalu v České republice – Designblok 2020. Autorka této práce navrhla a zrealizovala celou expozici. Díky této výstavě, kterou navštívilo nespočet lidí, získala autorka velmi cenné informace a podněty k dalšímu výzkumu.

11. List of abbreviations

AS – Acid sulfite

CB – Cardboard

DIP – Deinked pulp

FDM – Fused Deposition Modeling

FFWT – Faculty of Forestry and Wood Technology

FLEC - Field and laboratory emission cell

OCC – Old corrugated cardboard

OP – Office paper

HDPE – High-Density Polyethylene

HEM – High-Energy Milling

IB – Internal Bond

ICWSE – International Conference *“Wood Science and Engineering in the third millennium”*

IGA – Internal Grant Agency

INCIEN – Institute of Circular Economy (Institut cirkulární ekonomiky)

LCA – Life-Cycle Assessment

LSU – Louisiana State University

MENDELU – Mendel University in Brno

MDF – Medium Density Fiberboard

MOR – Modulus of Rupture

MOE – Modulus of Elasticity

PET - Polyethylene terephthalate

PP – Polypropylene

PLA – Polylactic acid

PVC – Polyvinylchloride

PVAc - Polyvinyl acetate

SLA – Stereolithography

SUAS – Salzburg University of Applied Sciences

SWR – Screw Withdrawal Resistance

VOCs – Volatile Organic Compounds

TS – Thickness swelling

TVOCs – Total Volatile Organic Compounds

WA – Water Absorption

WHO – World Health Organization

WPC – wood plastic composite

UF – Urea-formaldehyde

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